

U.S. Cities' Renewable Energy Targets

Rhetoric or Reality?

April 4, 2019

Environmental Studies Thesis

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Acknowledgements

Special thanks to my family for all of their support, the faculty in the Environmental Studies Department and the Energy Studies Program, and Nathan Serota, for teaching me nearly everything I know about the energy landscape and mentoring me through much of this research.

Glossary

Table 1: Terms and definitions

Term	Definition
Behind-the-meter	Electric generation in the case that the generating unit is located in the same site as the electric off-taker
City-utility franchise agreement	A legal contract that outlines requirements for a utility to use the city's public rights of way. This typically involves compensation paid to the city via a pass-through charge to ratepayers.
Community choice aggregation (CCA)	CCAs (also referred to as municipal aggregation) enable local governments to aggregate the electricity loads of residents, businesses, and/or municipal facilities. This state policy allows municipalities to select an electric provider on behalf of their residents, businesses, and municipal accounts.
Community Solar	A solar-specific variation of community shared renewables.
Community-shared renewables	A renewable energy facility that allows multiple people to subscribe and benefit from the energy output of the facility. Shared renewable energy projects often typically utilize a special tariff/rate, similar to aggregate or virtual net metering policies, that facilitate the utility's metering and billing of participating customers.
Cooperative Utility	Energy co-ops are non-profit institutions run by the customers that the cooperative serves. They are formed to help reduce individual cost for products and/or services and share expenses among a group of customers. Each customer is a member-owner with one vote under the "one person, one vote" cooperative principle. Cooperative Utilities are governed by a board of directors, with only minimal oversight from state regulators.
Decoupling	A state-level regulation that separates utility profits from the volume of direct energy sales. Utility revenue is instead predetermined by regulators, often based on recovery of fixed costs, and performance, and customer tariffs are set to meet this target. This regulation is designed to remove utility disincentives to adopting energy efficiency programs and other improvements that reduce customer demand.
Deregulated utilities	Utilities operating in a marketplace with retail competition. Many deregulated utilities provide either generation or transmission or distribution services, as opposed to Regulated Utilities, which provide all three services in their territory. Markets may be deregulated or restructured at two different levels: • Deregulated at the wholesale level – In regions where the electricity market is deregulated at the wholesale level, utilities maintain a monopoly over transmission and distribution but not generation; competition exists at the wholesale level, and independent system operators (ISOs) or regional transmission organizations (RTOs) manage the market (including generation scheduling and pricing). Physical and virtual PPAs may be signed in deregulated wholesale markets. • Deregulated at the retail level – In regions where the electricity market is deregulated at the retail level, utilities maintain ownership and operation of transmission and distribution (as with wholesale deregulation). However, customers have the option to choose alternative retail electricity suppliers. This allows for retail competition in regions managed by ISOs or RTOs. Retail level deregulation permits contracting directly with independent power producers (IPPs) to source renewable energy.
Direct Access Tariffs	Direct access tariffs allow large energy users in traditionally regulated energy markets to choose a supplier other than the distribution utility. As with a PPA and unlike a Utility Renewable Tariff, Direct Access Tariffs allow customers to purchase power directly from

	independent renewable energy suppliers, with the utility continuing to provide transmission and distribution services.
Distributed energy generation	Small-scale generation resources that feed directly into distribution portions of the electric grid (e.g., rooftop solar).
Green Power Purchasing Program	Green tariff or green tariff-like structure (such as a sleeve PPA). In vertically-integrated states where customers have only one utility option, the utility may offer the option to add a premium per unit of energy to a customer's electricity bill for renewable energy (also known as a "green power" option) that consists only of RECs.
Green pricing program (GPP)	A retail energy offering made by utilities to customers in which a voluntary premium is added to energy bills to reflect the environmental assets of renewable energy generation.
Green tariff	A green tariff (also known as a utility renewable tariff) is a special utility-provided renewable energy purchasing option. In vertically-integrated states where customers have only one utility option and cannot directly negotiate PPAs, the utility may offer a green tariff program that include both an electricity price and RECs as an option for customers.
Home rule state	Whether or not a city is located in a Home Rule state dictates the level of governing authority granted to municipalities. Cities in Home Rule states have more power over local decision-making (such as passing energy ordinances) than those in Dillon's rule states. Cases in which individual larger cities within Dillon's Rule states are granted Home Rule authority are also noted.
Independent power producer (IPP)	A company that generates and sells power.
Independent system operator	for cities located in regions with organized wholesale markets the independent system operator runs the network and facilities comprising the transmission grid.
Investor-owned utility	A utility owned by private investors, as opposed to one owned by a public trust or agency. A commercial, for-profit utility as opposed to a co-op or municipal utility. Issues company stock and is responsible to shareholders. Heavily regulated at the state level.
Municipal Utility	Utility that is owned by a municipal government.
Net energy metering (NEM)	NEM allows customers who own renewable assets on site (such as rooftop solar) to sell excess renewable generation back into the grid and be compensated either through a credit to, or reduction on, their monthly utility bill – thereby improving the financials for renewable investment.
Power purchase agreement (PPA)	A contract to buy electricity from a generator that sets a term (usually 10-25 years) and an electricity price (in terms of \$/kWh). There are multiple PPA structures, the two most popular being physical PPAs and virtual PPAs.
Public Service Commission	A state agency that regulates the rates and services charged and provided by intrastate utilities.
Public Utilities Commission	A state agency that regulates the electric utilities in a given state.
Regulated utilities	Regulated utilities exist in states whose electric utilities are vertically-integrated. This means that generation, transmission, and distribution are all performed by the same utility, and cannot easily be decoupled. Utilities in these territories maintain a natural monopoly over all three stages. While utilities may choose to contract with other generators, customers who purchase from the utility in these areas cannot specify how their electricity is generated. Physical and virtual power purchase agreements (PPAs) are not applicable in regulated markets. However, deal structures with the local utility such as green tariffs and green power purchases may exist.
Renewable energy certificate	Cities can purchase and retire renewable energy credits (RECs) that are generated from offsite renewable projects. The terms unbundled and bundled denotes whether RECs purchased come alongside clean energy in their contracts (bundled) or if the environmental attribute is sold as a separate (unbundled) entity.

Renewable energy targets	See municipal renewable energy target and community wide renewable energy target. This analysis categorizes targets as 'Active' or 'Achieved'. (1) Active renewable energy targets: Captures 61 cities that have made renewable commitments with deadlines in the foreseeable future. Cities with populations below 5,000 are excluded from this section for purposes of scale. (2) Achieved renewable energy targets: Details the successes of 18 entities. A population minimum is not imposed in this list so as to encompass the range of strategies employed.
Renewable energy target: Community-wide renewable target	The target percent of renewables to be incorporated into the electricity/energy supply for the entire community's operations (residential, commercial, industrial, municipal, etc.) based on public commitments.
Renewable energy target: Municipal renewable target	The target percent of renewables to be incorporated into the electricity/energy supply for a city's municipal operations only (city owned buildings, government facilities, etc.) based on public commitments.
Renewable portfolio standard	A policy (typically state level) mandating that a certain percentage of electricity provided by the utility must be generated by renewable resources by a target year.
Retail choice	Gives the customer has a say in how their electricity is generated also known as 'retail competition' or 'retail electric choice'. In states with retail choice customers are able to purchase their electricity from a retail electric supplier other than the local utility. In such 'deregulated retail' markets energy customers may have several provider options some of which could source renewables either through contracts with renewable generators or through the purchase of renewable energy certificates (RECs).

Source: For the purposes of directness and not re-writing already succinct and effective industry standard definitions, many of the above definitions are pulled from the World Resource Institute's *Implementation Guide for Utilities*¹ and Meister Consultants Group's *Pathways to 100* report.²

Table 2: Frequently used acronyms and abbreviations

Abbreviation	Description
CAISO	California Independent System Operator
CAP	Climate action plan
CCA	Community Choice Aggregator
CNCA	Carbon Neutral Cities Alliance
Coop	Cooperative Utility
DSIRE	Database of State Incentives for Renewables & Efficiency
EDF	Environmental Defence Fund
Elec	Electricity
EPA	Environmental Protection Agency
ERCOT	Electric Reliability Council of Texas
FERC	Federal Energy Regulatory Commission

¹ Priya Barua. (2017). Implementation guide for utilities: designing renewable energy products to meet large energy customer needs. WRI (World Resources Institute).
http://www.wri.org/sites/default/files/Implementation_RenewableEnergy_0.pdf

² Meister Consultants Group. (2017). Pathways to 100: Energy Supply Transformation Primer for Cities.
http://www.mc-group.com/wp-content/uploads/2017/08/MCG-Pathways-to-100_Energy-Supply-Transformation-Primer-for-Cities.pdf

Abbreviation	Description
FPL	Florida Power & Light
GP	Green power
IOU	Investor-owned utility
IPP	Independent power producer
ISONE	Independent System Operator of New England
kWh	kilowatt-hour
MISO	Midwest Independent System Operator
MWh	Megawatt-hour
NEM	Net energy metering
NREL	National Renewable Energy Laboratory
PG&E	Pacific Gas and Electric
PJM	Pennsylvania, Jersey Maryland power pool
PPA	Power purchase agreement
PSC	Public Service Commission
PSCo	Public Service Co of Colorado
PUC	Public Utilities Commission
PV	Solar
PV	photovoltaic
RE	Renewable energy
REC	Renewable energy credit
Retailer	Retail electric supplier
RFI	Request for information
RFP	Request for proposal
RFQ	Request for quote
RMP	Rocky Mountain Power
RPS	Renewable portfolio standard
RTO/ISO	Regional transmission organization / Independent system operator
SPP	Southwest Power Pool
TWh	Terra-watt hour

Section 1. Abstract

93

Cities captured in this analysis

18

Cities have achieved their renewable energy targets

14

Progressive cities are contracting renewable electricity regardless of any targets

61

Cities captured in this analysis are actively pursuing renewable energy targets

More than 61 U.S. cities have announced seemingly ambitious renewable energy commitments. Are the headlines just noise, or could they offer a significant new source of demand for renewable energy developers and their project investors?

Cities have been making – and in some cases, meeting – renewable energy commitments for years, but the frequency of new announcements and the attention their gaining in the public sphere has increased following the 2016 election.³ In the absence of legal coercion or clear economic incentives these cities' behaviours represent a highly unusual example of volunteerism. Regardless of the motives driving each individual city, the result is significant new demand for renewables spurring out of a popular movement to reduce each communities' influence on the planet. To get a realistic sense for the market opportunity created by the recent surge in municipal renewable energy commitments, this study analyzes the demand profiles of 93 U.S. cities: 61 actively pursuing targets, 18 that have achieved their targets, and 14 cities that are procuring renewable electricity as if they had formal targets. The goal of this analysis is to quantify the amount of new demand for renewable energy at the city level driven by the 61 cities actively pursuing commitments with deadlines in the foreseeable future. Accordingly, this paper also examines how cities plan to procure renewables in compliance with their targets – whether it be through on-site generation, offsite or virtual power purchase agreements, or less direct methods such as purchasing renewable energy credits. The options available to cities for achieving their renewable energy commitments are highly dependent on their utility type and the surrounding regulatory landscape.

A note on scope: This note captures the universe of active and formally adopted commitments to substantial city-level renewable energy procurement across the U.S. and quantifies the incremental demand for new renewables driven by these targets.

There are different types of city-level climate and energy commitments (i.e., greenhouse gas emissions reductions, renewable energy, renewable electricity, etc.). This research focuses exclusively on cities making ambitious renewable electricity targets (>10% or higher) for either their entire community or their municipal operations alone. The data encompasses all city commitments announced as of **April 2018**.

A note on sourcing: The results presented in this research are based on detailed data collection, a series of interviews, and original analysis. Please see the accompanying excel documents for quantitative and qualitative city-specific raw data⁴. Appendix B contains more details on sourcing as well as guidance on navigating the excel files and the incorporated data visualization tools. Appendix C includes two tables summarizing some of the high-level data presented in the excel

³ See Figure 1 for a timeline of renewable energy commitment announcements.

⁴ Two excel files accompany this research – the first is a descriptive dataset containing collected raw data and bibliographic links for each city's target, overall approach, electricity profile, and relevant regulatory information. The second file contains the results and demand analysis upon which all of the following forecast conclusions presented in this paper are based. This 'Results and underlying analysis' file also includes two data visualization tools: one for toggling by regulatory and geographic factors, and another for toggling the demand forecast by selected city. (See Appendix B for a description the tabs in each file).

files in order to provide a more concise overview of each cities' electric profile, renewable strategy, and documented actions to date.

61

'Active' targets

40

Active targets set to meet 100% community-wide electricity demand with renewables

35

Commitments announced in 2017 alone

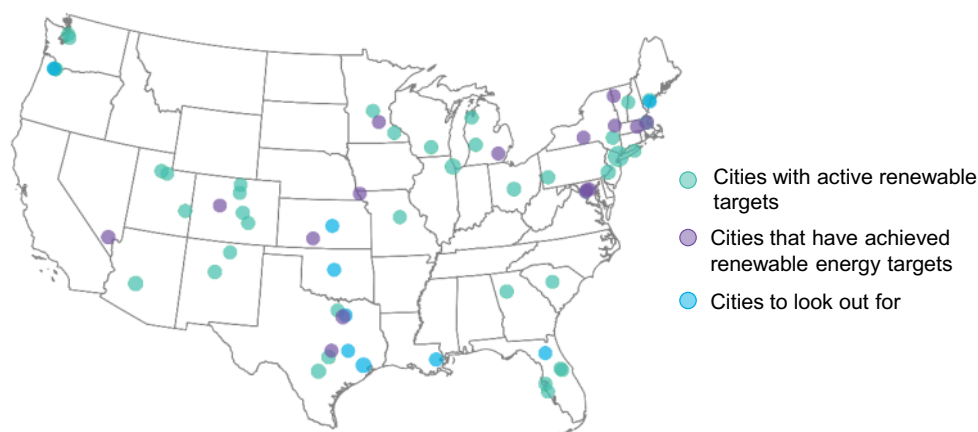
74TWh

Of unmet demand for municipal wind and solar electricity that must come online by 2035 if all 61 cities are to meet their commitments

Key findings:

- 8 of the top 10 largest U.S. cities by population have renewable energy targets (four of which are for 100% community wide operations), and the other two of the 10 most populated U.S. cities are procuring renewables actively enough to fall in this studies' list of 'looming cities.'⁵
- The total annual renewable electricity needed for these cities to achieve their commitments amounts to 103TWh in 2035. After accounting for the portion of this demand already spoken for by state renewable portfolio standards (RPS), existing and contracted supply, and expected renewable energy credit (REC) purchases, the analysis forecasts the majority of this demand (74TWh) to remain unmet come 2035 – thereby representing a potential market opportunity for renewable developers and possible hurdle for the affected utility service providers.
- Near term demand will be driven by cities with flexible renewable procurement options – i.e., cities with retail choice or that have their own municipal or rural cooperative utility. Come 2035 however, just under half the municipal demand for renewables (42%) will fall in vertically integrated utility territories without retail choice.
- As with corporations working towards renewable targets, and utilities complying with renewable portfolio standards, there is a range of strategies that cities may implement in meeting their goals. Significant ambiguity still surrounds the level of ambition with which cities will strive to achieve their targets.

Cities included in this dataset colored by renewable energy target status



Source: Map generated in QGIS based on data collected from interviews, various city and climate action public announcements, the Bloomberg Terminal, and the U.S. Census Bureau. Notes: This map excludes all cities captured in California and small populations (<5,000 residents). Methodology of inclusion and categories are described in Table 7 of Appendix B.

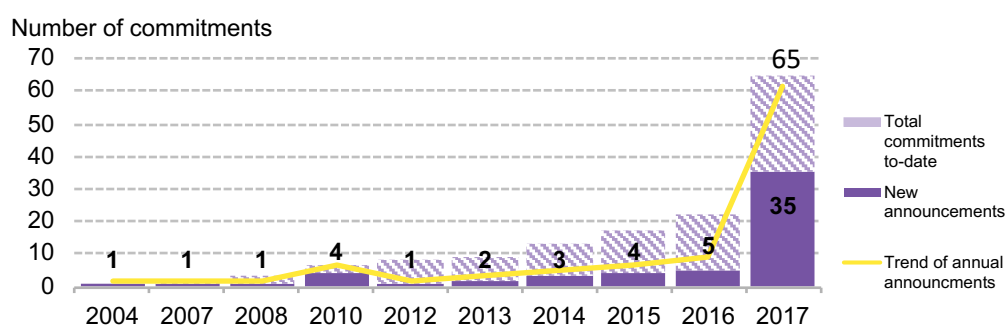
⁵ See the attached U.S. Cities Renewable Energy Targets and Strategies Dataset for a list of all cities that have active targets, achieved targets and are considered to be looming. New York, Chicago, Philadelphia, Phoenix, San Antonio, San Diego, and San Jose all have 'active' RE targets with deadlines in the foreseeable future. Dallas has an achieved municipal level target, and Los Angeles and Houston are both actively procuring renewable electricity; thus their profiles are detailed on the 'looming' tab.

Section 2. Background

2.1. Recent trends

U.S. cities have been making – and in some cases, meeting – renewable energy commitments since at least 2004 when Aspen Colorado established the earliest documented meaningful city target of becoming a 100% renewable community by 2015.^{6, 7} Between 2004 and 2016, 44 cities⁸ announced renewable energy goals, equivalent to under four per year. Over the last twelve months, however, the number of fresh commitments has increased dramatically, with 35 new announcements made in 2017 alone – over seven times the number in 2016.

Figure 1: Timeline of active city renewable energy commitment announcements



Note: Excludes six cities with ambiguous announcement dates. Detailed methodology, the criteria for inclusion, and assumptions can be found in Appendix A.1. See Appendix C for a full list of the cities included in this section's analysis. The raw data collected and used to propagate this chart is available in the accompanying 'U.S. City Renewable Energy Targets and Strategies' dataset.

This is part of a trend: inspired by private-sector companies' adoption of renewable energy goals⁹ and motivated by activist efforts like America's Pledge, cities are increasingly taking renewable energy procurement into their own hands – whether it be to reduce electricity rates, to better their local environments, or to make a political statement – and they're often saving on their electricity

⁶ National Renewable Energy Laboratory. (2015). Reaching 100% Renewable Energy: City of Aspen and NREL Implementation Strategy. *NREL.gov*. <https://www.nrel.gov/docs/fy15osti/62490.pdf>.

⁷ Aspen achieved this commitment by negotiating with both the Municipal Energy Agency of Nebraska and the Holy Cross Energy local cooperative to increase their renewable supply and to include renewable resource requirements in the utility franchise agreements. Aspen's current energy portfolio is about 50% wind, 45% hydropower, and 5% solar and landfill gas. See the accompanying excel dataset or Table 13 within Appendix C.2 for a full list of cities that have achieved their renewable energy targets.

⁸ The term 'city' is used throughout this document and in the corresponding spreadsheet to refer to the various entity types covered by this analysis: cities, towns, counties, boroughs, and municipalities. There are different types of city-level climate and energy commitments (e.g., greenhouse gas emissions reductions, renewable energy, renewable electricity, etc.). This note focuses exclusively on cities making ambitious renewable electricity targets (>10% or higher).

⁹ For an in-depth look at the rise in private-sector renewable energy procurement in the U.S., see Bloomberg New Energy Finance's October 25, 2016 Research Note, *What's the growth opportunity for US corporate PPAs?* ([web](#) | [terminal](#)); and for an updated, comprehensive list of related research, see BNEF's *Corporate Procurement* [theme page](#).

bills as a result. Despite the U.S. Federal Governments' stance, 238¹⁰ cities have committed to the America's Pledge—an effort to spur sub-national commitment to the Paris Agreement goals and climate actions at large—and over 350¹¹ cities have signed on to similar campaigns such as the U.S. Climate mayor's compact, Go100RE,¹² and Sierra club's cities for 100% renewable energy.¹³ Today, 79 cities have official renewable energy targets on the books and at least 14 others are actively purchasing renewable energy as if they did.¹⁴ 18 cities have achieved their targets, while the remaining majority of these cities (61 of the 79) have 'active' commitments – i.e., renewable electricity targets with deadlines in the foreseeable future. As creditworthy electricity users with long-term time horizons, they therefore represent an attractive source of demand for renewable energy developers and utilities.

2.2. U.S. Cities' Renewable energy targets

Two types of city renewable energy targets:

City renewable energy targets generally fall into one of two categories.

1. The commitments are for cities' *municipal operations* only, targeting renewable energy in order to supply the electric load of city government buildings and facilities.
2. The commitments are for *community-wide operations*, which encapsulate the entirety of the city's electricity demand – not just for the city's own operations but also for the citizens and businesses within the city.

In some cases, cities commit first to a target covering their municipal operations as an interim step toward a target covering their community-wide operations.¹⁵

¹⁰ See the November 2017 [America's Pledge Phase 1 Report](#) for more information on the joint initiative between Michael Bloomberg and Jerry Brown to assess non-federal U.S. climate actions and the We Are Still In coalition.

¹¹ Number compiled as of November 2018 from data available on each campaign's website.

¹² Renewables 100% Policy Institute. (2018). Go 100% Renewable Energy: Projects in North America. *Go100percent.org*. <http://www.go100percent.org/cms/index.php?id=18>.

¹³ Sierra Club. (2017). Ready for 100 commitments. <https://www.sierraclub.org/ready-for-100/commitments>.

¹⁴ For a full compilation of the cities that this analysis has been tracking in terms of their renewable energy initiatives, see the 'Targets' tab in the accompanying 'US Cities RE Targets – Results and Underlying Analysis' excel file #2.

¹⁵ The accompanying Excel dataset breaks out municipal operations targets separately from community-wide operations targets to highlight this difference.

Section 3. How cities buy electricity

Supplying customers with electricity involves three key stages: (1) generation, (2) transmission, and (3) distribution. Whether or not a particular electricity market is 'regulated' or 'deregulated' determines who can play a part in the final two stages, namely transmission and distribution.¹⁶ While multiple entities may be able to generate electricity, the regulatory structure determines who can contract electricity from such generation – i.e., if consumers can elect to buy from a particular generation source (an ability called 'retail choice') or if only industrial-owned utilities (IOUs) have the authority to do so, and customers must go through the utility to obtain their electricity.¹⁷ See the map in Figure 2 for a delineation of regulated and deregulated territories in the USA.

Regulation 101

- **Regulated territories** – these are states whose electric utilities are vertically-integrated. This means that generation, transmission, and distribution are all performed by the same utility, and cannot easily be decoupled.¹⁸ Utilities in these territories maintain a natural monopoly over all three stages. While utilities may choose to contract with other generators, customers who purchase from the utility in these areas cannot specify how their electricity is generated.¹⁹ Physical and virtual power purchase agreements (PPAs) are not applicable in regulated markets. However, deal structures with the local utility such as green tariffs and green power purchases may exist.

Deregulated Markets may be deregulated or restructured at two different levels:

- **Deregulated at the wholesale level** – In regions where the electricity market is deregulated at the wholesale level, utilities maintain a monopoly over transmission and distribution but *not* generation; competition exists at the wholesale level, and independent system operators (ISOs) or regional transmission organizations (RTOs) manage the market (including generation scheduling and pricing). Physical and virtual PPAs may be signed in deregulated wholesale markets.
- **Deregulated at the retail level** – In regions where the electricity market is deregulated at the retail level, utilities maintain ownership and operation of transmission and distribution (as with wholesale deregulation). However, customers have the option to choose alternative retail electricity suppliers. This allows for retail competition in regions managed by ISOs or RTOs. Retail level deregulation permits contracting directly with independent power producers (IPPs) to source renewable energy.²⁰

¹⁶ DOE. (January 2017). Transforming the Nation's Electricity System: The Second Installment of the QER. Appendix A. http://www.energy.gov/sites/prod/files/2017/01/f34/Appendix%20_0.pdf.

¹⁷ Meister Consultants Group. (2017). Pathways to 100: Energy Supply Transformation Primer for Cities. http://www.mc-group.com/wp-content/uploads/2017/08/MCG-Pathways-to-100_Energy-Supply-Transformation-Primer-for-Cities.pdf

¹⁸ World Wildlife Fund and The World Resources Institute. (2016). U.S. Renewable Energy Map: A guide for corporate buyers. REBA: Corporate Renewable Energy Buyers' Alliance Principles. <http://buyersprinciples.org/corporate-re-strategy-map/>.

¹⁹ Ellen Flynn Giles and Kathy L. Brown. (2014). UDI Directory of Electric Power Producers and Distributors: 123rd Edition of the Electrical World Directory. New York, NY: Platts.

²⁰ Law360. (July 29th 2009). Utilities and Renewable Energy: To Buy or to Build. Portfolio Media, Inc., 9.

Why does this matter to cities?**In regulated areas:**

- A city cannot cite physical or virtual power purchase agreements (PPAs)²¹
- Green tariff or green power purchase deal structures with local utilities may exist

In deregulated areas:

- Cities have more flexibility in how, and from whom, they can procure renewables as they are allowed to contract directly with independent power producers (IPPs) to source renewable energy.

Why does this matter to developers?

- Cities with renewable commitments that are located in areas with retail electric choice (i.e., where the customer has a say in how their electricity is generated) could offer a new source of demand and market opportunity for renewable energy developers and IPPs.

Policy and regulation – typically at the state-level – dictate how different cities buy their electricity and therefore how they buy (or build) renewables.

In regulated territories, cities must buy electricity via contracts with their local investor-owned utilities, as they monopolize the three stages of electricity supply, transmission, and distribution. In states with retail choice, the local utility maintains ownership and operation of transmission and distribution, but cities may choose to contract with alternative retail electric suppliers. In both cases, from the perspective of the electricity provider, in the absence of a municipal operation city governments are similar to other large commercial and industrial electricity customers.²²

Municipal, rural and cooperative utilities:
some city governments
operate their own utilities

Some cities operate their own utility.²³ In these cases, both the city government and all of the businesses and citizens within the city pay their respective electric bills to the same municipal utility. Not surprisingly, this affects their renewable strategy. Cities with their own utility operation or with retail choice (i.e., the ability to choose their initial generation source) typically have more flexibility than cities served by a single, vertically-integrated investor-owned utility.

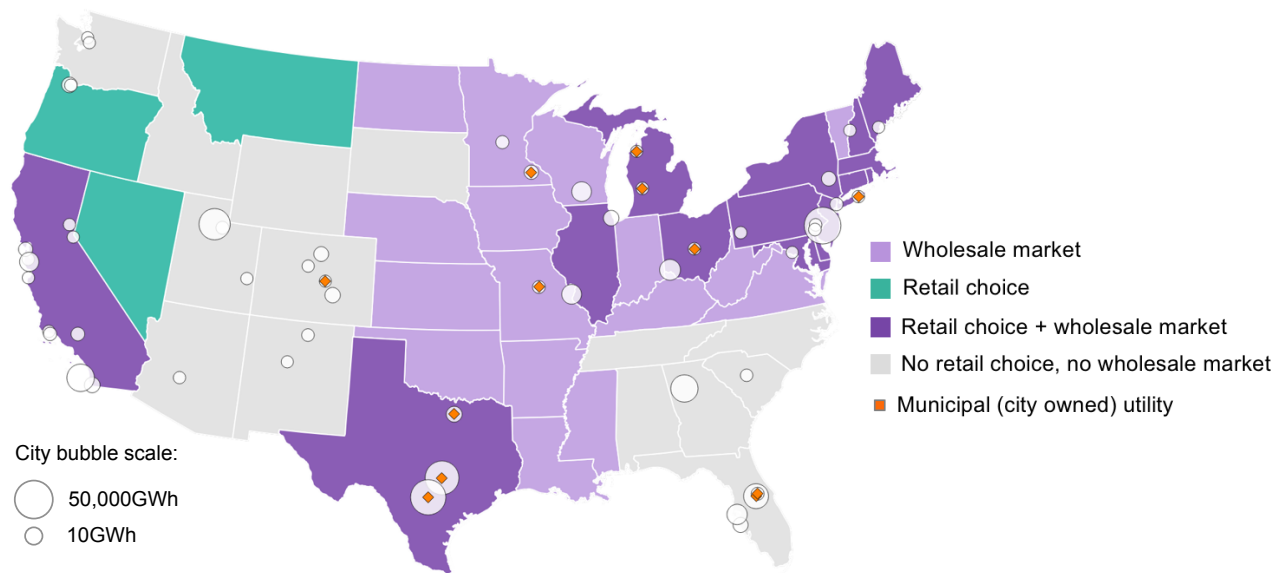
The map in Figure 2 depicts the regulatory landscape surrounding the 61 cities with active renewable targets captured in this analysis.

²¹ RMI. (2016). Increasing the Pace of Corporate Participation in New York's Renewable Energy Programs: A Proposal by Rocky Mountain Institute's Business Renewables Center. Confidential Draft Document.

²² For more on corporate renewable energy strategies see Bloomberg New Energy Finance's 2016 insight: Nicholas Albanese. (2016). What's the growth opportunity for US corporate PPAs?. <https://www.bnef.com/core/insights/15295>.

²³ There are 2,941 unique utilities in the U.S. of which 62% are municipal, and 28% are rural or cooperative entities. (Source: DOE Quadrennial Energy Review 2017 <https://www.energy.gov/sites/prod/files/2017/01/f34/Appendix%200.pdf>)

Figure 2: Map of states colored according to retail choice offering, with squares indicating relevant municipal service territories, and bubbles for each city renewable energy target, sized by estimated incremental demand in 2040



Source: Original graphic made in QGIS using data collected from interviews, publicly-available local government websites, publications, and announcements, as well as the Bloomberg Terminal. Electricity supplier information is consistent with annual data released by the Energy Information Administration (EIA), specifically *Form 861*.²⁴ See Appendix A for electricity profile modeling. Regulatory landscape information is consistent with the Federal Energy Regulatory Commission (FERC), and the Database of State Incentives for Renewables & Efficiency (DSIRE).²⁵

Notes: City bubbles are sized on a logarithmic scale. Note also that: cities served by municipal utilities are assumed to have no retail choice.

In areas with **retail choice but no wholesale market** (green) the local utility maintains ownership and operation of transmission and distribution, but cities may choose alternative electricity suppliers. In areas with **both a deregulated market and retail choice** (dark purple) cities can contract directly with independent power producers to source renewable energy. Physical and virtual PPAs may be signed in **deregulated wholesale markets** (light purple)²⁶. In **vertically integrated markets** (grey) cities must work with their local utility as neither form of PPAs are permitted. See Table 3 for further details on the various renewable electricity procurement options in different regulatory structures. See Appendix C for a full list of the cities include and the accompanying excel files for the raw data used to generate this graphic.

²⁴ Energy Information Administration. (2016). Form 861: The Annual Electric Power Industry Report. <https://www.eia.gov/electricity/data/eia861/>.

²⁵ DSIRE. (April 2017). Retail Choice. www.dsireusa.org.

²⁶ Hugh Bromley. (2017). Options Overload: The many varieties of U.S. commercial solar. Bloomberg New Energy Finance. Bloomberg L.P. Terminal, New York, NY.

3.1. How cities buy renewable electricity

There are a number of ways for cities to participate in the renewable energy industry to meet their targets.

- **Contracts:** Cities can procure renewables through a power purchase agreement (PPA), a contract with a generator that sets a term (usually 10-25 years) and an electricity price (in terms of \$/kWh). There are multiple PPA structures, the two most popular being physical PPAs and virtual PPAs.
- **RECs:** Cities can purchase and retire renewable energy credits (RECs) that are generated from offsite renewable projects.
- **Onsite generation:** either **behind-the-meter** or **net-metered** (allowing the city to sell any extra electricity that they generate back into the grid).
- Green tariff or green tariff-like structure (such as a sleeve PPA)

The local regulatory structures determine which of the options listed above are available to each city. Cities that possess retail choice, or alternatively, have their own municipal utility typically have more flexibility in procuring renewables compared to cities served by a single, vertically integrated investor-owned utility. Thus, in the near term, this analysis primarily expects to see progress on the targets set only for *municipal operations* (31 of the 61 commitments in total), and across *community-wide* commitments within the 13 cities that own their own municipal utility. There is less of a clear path to achieving community-wide electricity goals for cities that do not own their own utility.

In deregulated markets with retail choice the city government can contract directly with independent power producers (such as a wind or a solar developer) to source renewable energy. However, this renewable electricity may only be physically delivered to city-owned meters, not to the community as a whole.

In cities that own their own municipal utilities, the utility itself will contract with renewable developers on behalf of the entire city and deliver the electricity to the city government and all citizens and businesses within the community.

Table 3 summarizes how cities buy electricity in different landscapes, implying the implications of this landscape for sourcing renewables to meet either municipal or community-wide energy targets.

Table 3: How different cities buy electricity and the implications for sourcing renewables

Retail market structure	How a city buys electricity	How a city meets a municipal operations renewable energy target	How a city meets a community-wide renewable energy target
Municipal utility	All members of the city and the city government pay bills to the city's municipal utility on a regular basis.	• The municipal utility signs physical PPAs directly with renewable assets to purchase electricity on behalf of all the customers within the city. • <i>Options available: PPAs, voluntary REC purchases</i>	• <i>Same method as meeting municipal target.</i>
Deregulated market with retail choice	In a retail choice areas, the city government contracts with retail suppliers for its electric load. People who live in the city separately pay for electricity under retail supply contracts.	• City governments will work with different retail suppliers to sign offsite PPAs for renewable electricity; however, the electricity may only be physically delivered to city-owned meters. • <i>Options available: PPAs (with different retail suppliers), onsite generation (either net metered or fully behind the meter, pending state legislation), voluntary REC purchases</i>	• No set strategy for influencing from which retail provider the citizens and businesses choose to buy their electricity.
Vertically integrated utility	The city, its members, and often neighbouring communities, must all buy electricity under contracts with the single investor-owned utility covering their service area.	• Cities without the ability to purchase electricity directly from suppliers must work with their local utility to pursue a contract either through a dedicated utility provided renewables program such as a green power purchase program, a green tariff program, or green tariff-like arrangements (otherwise known as a sleeve PPA). • <i>Options available: Onsite generation (either net metered or fully behind the meter, pending state legislation), voluntary REC purchases, green tariff programs, green power purchase programs</i> <i>Example: The Energy Rider program in North Carolina offers PPA-like options for large customers such as municipalities to source renewable electricity through Duke Energy, their local IOU.</i>	• No set strategy for influencing the energy mix that the vertically integrated utility provides to the citizens and businesses in the city. Cities must either work with, or in some cases separate from, their local utility. <i>Note: Some of the more creative strategies that different cities in this position are pursuing include leveraging buying power (particularly for those cities who are key utility accounts) to renegotiate franchise agreements or request changes to the energy mix, leveraging taxing authority, engaging in state public utility proceedings, and seeking to municipalize the local grid through purchasing the privately-owned existing utility.²⁷ See Section 5 for further discussion of city's approaches.</i> <i>Example: Boulder, CO has been working to separate from their investor owned utility (PSCo) and municipalize since 2010.</i>

²⁷ Meister Consultants Group. (2017). Pathways to 100: Energy Supply Transformation Primer for Cities.

Section 4. U.S. city renewable energy demand forecast

This section quantifies the incremental demand for new renewable electricity at the city-level driven by 'active' city renewable energy targets. More specifically the analysis focuses on separating out the unmet or 'net' demand that has yet to be accounted for and thus represents an opportunity for new development.

Cities captured in this section

While this study tracks a total of 93 cities that have been participating in some way or another within the renewables area, the following demand forecast focuses only on cities with formal and unmet commitments. This section analyzes the subset of 61 cities actively pursuing renewable energy targets as defined in Appendix A. Each city captured has a population of at least 5,000 and has formally adopted a renewable target within their city council or government.

Methodology in a nutshell

To forecast U.S. cities' remaining (or unmet) net demand for renewable electricity this analysis:

1. Identifies each cities' total electricity demand, using disclosed values where available;²⁸
2. Based on stated targets and deadlines, estimates how much of this will come from renewables;
3. Adjusts this amount to determine cities' current renewable energy demand *after* accounting for voluntary REC purchases,²⁹ state RPS compliance,³⁰ and existing power contracts – items which reduce cities' appetite for signing new long-term renewable energy contracts.

Detailed methodology and assumptions can be found in Appendix A. See Appendix C for a full list of the cities included in this section's analysis and the criteria for inclusion and see the accompanying 'US Cities RE Targets – Results and Underlying Analysis' excel file 2' for a direct look at the original analysis and supporting data.

Total electricity demand

The 61 cities with unmet renewable energy targets (described in Section 2 and detailed in Appendix C) currently purchase 1,892TWh each year, representing approximately 46% of U.S. electricity consumption. Looking forward through 2035, this analysis assumes annual electricity demand for each city grows at the same rate as the 2010-2015 electricity sales compound annual growth rate (CAGR) of their respective primary utility provider. Based on city electricity use trends over these years, the forecast suggests these 61 cities will use 6% less collective electricity by the

Aggregate renewable demand will amount to about 30TWh in 2025 and 103TWh in 2035...

²⁸ Data comes either from interviews conducted by the author, Miela Mayer, during 2017, publicly available municipal government publications, or it is calculated based on EIA Form 861 electricity supply data. See Appendix B.1 for more details on the raw data used in the accompanying excel analysis.

²⁹ Environmental Protection Agency. (2018). EPA Green Power Partnership: Top 30 local government. <https://www.epa.gov/greenpower/green-power-partnership-top-30-local-government>.

³⁰ Joshua Daniel. (November 14th 2017). U.S. Renewable Portfolio Standard (RPS) Demand Database. Bloomberg New Energy Finance. Bloomberg L.P., New York, NY.

close of 2035, while simultaneously increasing the portion of this total electricity that is committed to renewable sources.

Renewable electricity demand

...The unmet portion of this demand that may have implications for a developer is around 18TWh in 2025 and 74TWh 2035

By 2035, these cities will need 103TWh of renewable electricity to make good on their commitments (and 30TWh to do so in 2025). While about 40% of this total will be covered by state RPS, voluntary REC purchases and existing contracts³¹ (as shown in Figure 4); the remaining 74TWh (in 2035; 9TWh in 2025) of unmet demand will have to come from new projects. Assuming 60% is met with new wind build and 40% with solar, this municipal demand would require 27GW of new renewable capacity to be online by 2035³².

Figure 3: U.S. cities' collective annual electricity demand and the portion covered by stated renewable targets*

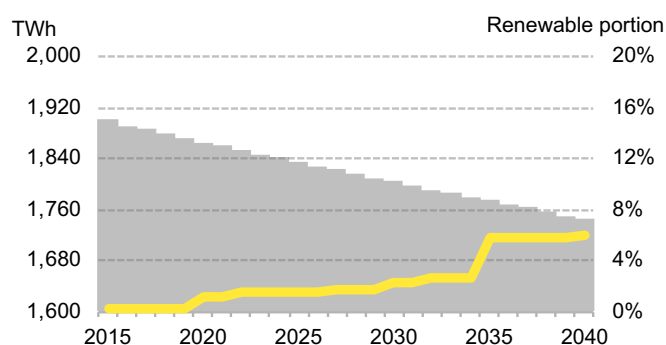
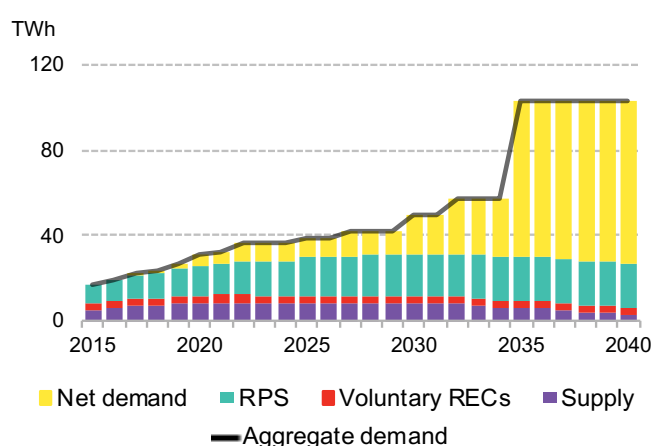


Figure 4: U.S. cities' renewable electricity demand, aggregate and net, based on stated targets*



Source: Notes: (*) Includes 61 cities with 'active' targets as described in Appendix A. In Figure 4, the total amount of 'targeted' renewable energy from city commitments is represented by the black aggregate demand line. The unmet portion of this demand that offers new opportunities for renewable development is represented by the yellow portion of net demand. See demand visualization tools in the accompanying results and analysis excel file to toggle this graph by city, regulatory landscape, and other relevant criteria

Figure 3 demonstrates the total electricity demand across the 61 cities as well as the increasing percentage of demand that is spoken for by their renewable energy commitments in each year through 2040. Figure 4 further breaks down this aggregate renewable electricity demand from the former figure, separating out the amounts that are already met by existing or contracted supply, effectively met through state RPS, and likely to be met with voluntary REC purchases, from the remaining portion of net (unmet) demand (represented in yellow).

³¹ State RPS, voluntary REC purchases and existing contracts will respectively cover 28%, 4% and 8% of demand in 2035. The amount of renewable supply estimated to be online decreases over time with the expected expiration of existing contracts, meanwhile REC purchasing trends are assumed constant. See Appendix A.2 for further details on the methodology and assumptions used for estimating cities' incremental demand for renewable electricity.

³² Calculations use an estimated capacity factor of 35% for wind and 27% for solar based on U.S. EIA 2016 annual averages.

Although the 6% renewable portion of these cities' collective electricity use is not striking as a standalone figure, it should be noted that the graph in Figure 3 shows the total community-wide electricity demand of all 61 cities. As mentioned earlier, there are two types of targets: community-wide targets and targets for municipal operations. The above shows the portion of total electricity demand (i.e. community-wide) that would need to be renewable to meet all city targets (both for municipal and community-wide operations). The fact that few large cities happen to have solely municipal-level targets—New York and Chicago for instance—yet their city-wide electricity demand is accounted for in the total, may dwarf the underlying trend that the majority of targets (40/61) are set for 100% of community-wide operations.

In some cases, the way that a city structures their community-wide target is by first setting a goal for transitioning some or all of municipal operations to renewable supply as a benchmark before transitioning all city-wide demand.

In fact, only 13 of the 61 cities have set targets for municipal-level operations alone and the net demand incentivized by these municipal-level targets is nearly insignificant (0.2%). Meanwhile 48 cities are pursuing community-wide targets, 18 of which have set interim municipal targets as well. If we look specifically at comparing the total electricity demand within these 48 cities with community-wide targets to their respective renewable demands, this percentage of the renewable portion jumps drastically from the 6% shown in Figure 3 to accounting for 78% of their collective electricity demand. Meanwhile the demand total electricity demand from the 13 cities with municipal-level-only targets accounts for a significant 87% of the collective electricity demand of *all* 61 cities included in this subset. What this means is that some of the biggest cities in this subset, and thus in the U.S., are the ones making the smallest commitments, which underrepresents the fact that the majority of cities captured are making significant commitments to transitioning their electric portfolios to renewables.

Another qualifier to keep in mind with Figure 3 is the depicted decline in forecasted collective electricity demand. It is important to consider that this decreasing electricity trend is very slight and easily reversed with increasing shifts toward electrification.

Evolving electric landscapes could impact demand

The trend in electrifying building energy use and shifting to electric vehicle fleets, particularly amongst those cities that have set additional climate or carbon neutrality targets (such as Phoenix, Chicago, and Salt Lake City) could drastically increase the total electric loads that cities face and thus their demand for renewable electricity – a situation Park City is encountering now as they shift their 50-bus fleet to being entirely electric.³³

4.1. Analysis by city

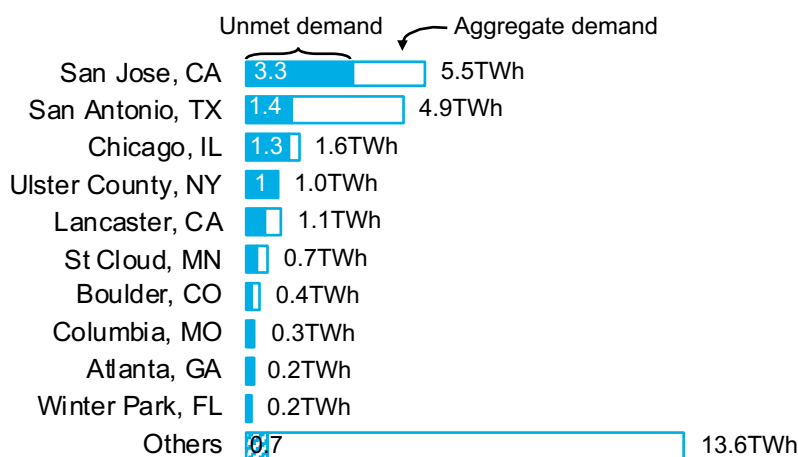
The unequal distribution of net and aggregate demand across different cities is demonstrated by the two graphs in Figure 4 — highlighting the top 10 cities in terms of unmet near-term (2025) and long term (2035) demand for renewable electricity. In the near term (2025) San Jose will account for 35% of net demand with 3.3TWh needed to meet their target. Without the amount that's already been accounted for by California state RPS,³⁴ this would be an even greater aggregate demand of 5.5TWh.

³³ Park City Government Source. (2017) July 19th, 2017 Interview. Interviewer: Miela Mayer.

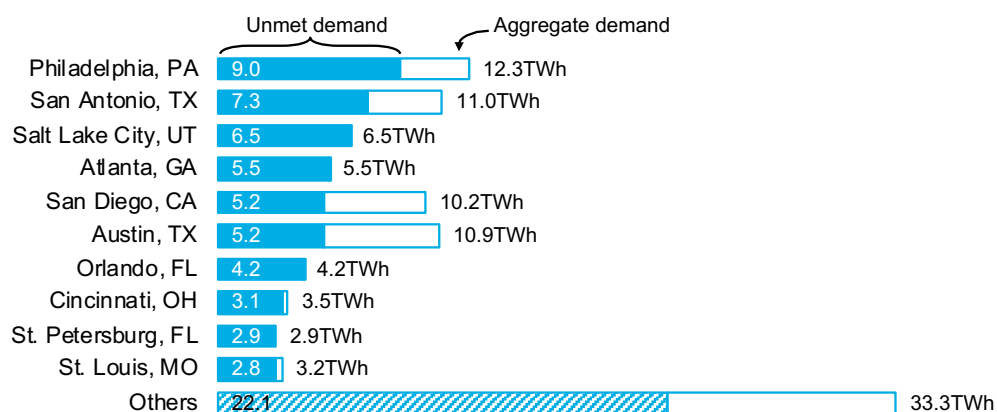
³⁴ Bloomberg New Energy Finance. (2016). 2016 California RPS Outlook: Eyes on the IOUs. Bloomberg L.P. Terminal, New York, NY. <https://bloom.bg/2egCbog>.

Come 2035, just six of the 61 cities analysed account for over 53% of this unmet demand: Philadelphia, PA, San Antonio, TX, Salt Lake City, UT, Atlanta, GA, San Diego, CA and Austin, TX – four of which rank among the 10 most populated cities in this research.

Figure 4: Top 10 U.S. cities ranked by annual *unmet* renewable electricity demand, 2025



Top 10 U.S. cities ranked by annual unmet renewable electricity demand, 2035



Overall the majority of this unmet demand (91% between 2020 and 2035) will be a result of community-wide energy targets. However, a few large cities with targets for municipal operations alone will still drive a significant portion in the near term. Most notably, Chicago and Ulster County (New York) come in third and fourth, respectively, when ranked in terms of unmet demand in 2025 (see Figure 4). As discussed above, such demand from municipal targets implies more immediate opportunities for development as city governments have a much clearer ability to start sourcing renewables for just the city owned buildings than cities seeking to cover their entire communities demand.

Case study on Chicago: Chicago, served by Commonwealth Edison Co (an industrial-owned utility) but located in an area with retail choice, is one of the few (18 of the 61) cities to have set an interim commitment, demonstrating intentionality in planning for meeting their overall commitment. The city has expressed intentions to sign long-term PPAs with retail suppliers, increase onsite generation, and purchase short term RECs generated under the Illinois RPS (25%

by 2025) to meet 50% of the electric load of about 900 municipally owned buildings by 2020, and 100% by 2025.³⁵

4.2. Analysis by region

Figure 5 shows the progression of net renewable demand from five of the aforementioned top 10 cities in 2035 on an annual basis through 2040, along with the demand driven by the 56 other cities captured. As is visible, the only city creating near term net demand is San Antonio, TX (yellow). San Antonio's early demand is due to a 20% community wide renewable electricity by 2020 interim target that the city chose to set as a first step to meeting their final goal of being a 40% renewable city. While this analysis assumes San Antonio's renewables will be purchased by 2035, the city's internal deadline is not till 2040. Having a municipal utility gives the city a direct path forward to procuring renewables. San Antonio has actually already contracted 8 solar power purchase agreements (signed between 2012 and 2015, all online as of 2017) through their municipal utility bringing them to an 11% renewable portfolio as of 2018.³⁶

Figure 5: U.S. cities' annual net renewable electricity demand by city

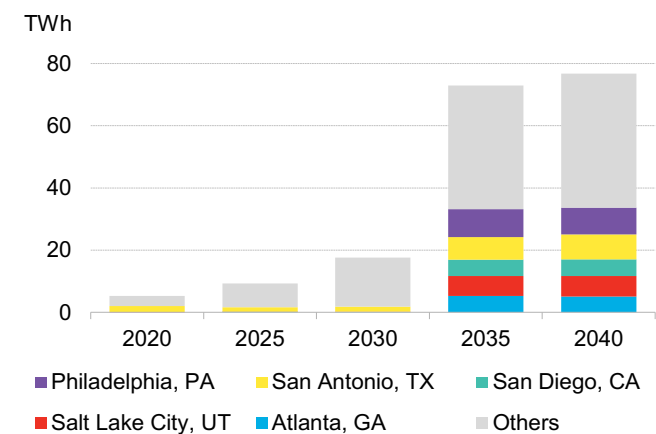
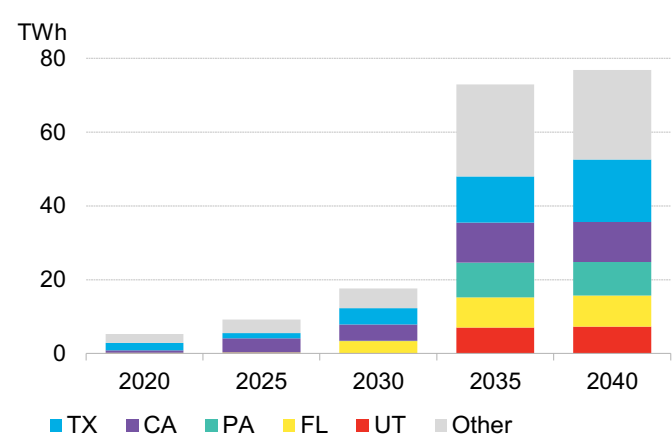


Figure 6: U.S. cities' annual net renewable electricity demand by state and region



City energy targets will drive new demand for renewables across 23 different states. However, as depicted in Figure 7, over half of this demand (66%) is concentrated in just 5 top states – Texas (20%), California (15%), Pennsylvania (15%), and Florida (13%). Near term demand in particular is highly concentrated in California with the state comprising 42% of all net demand in 2025. Come 2035, city energy deadlines that are more equally spread across other states will kick in, evening out the demand profile. Of the states carrying the majority of demand, Texas, California and Pennsylvania are deregulated areas offering many enabling policies for renewable energy

³⁵ Robert Walton. (April 2017). Chicago to power city-owned buildings with 100% renewable energy. Utility Dive. <https://www.utilitydive.com/news/chicago-to-power-city-owned-buildings-with-100-renewable-energy/440203/>

³⁶ See the 'supply' tab in the accompanying 'US Cities RE Targets – Results and Underlying Analysis' excel file #2 for more information on these 8 Solar PV contracts.

(such as retail choice, net metering, and community choice aggregation in California). However, Florida’s vertically integrated regulatory landscape is far less conducive to municipal renewable energy procurement. See Figure 8 for a further depiction of the trend towards increasingly significant demand come 2040 in areas with predominantly regulated utility markets. Municipalities will not be the only customers putting new demand on the regulated utilities for renewables. Bloomberg New Energy Finance’s analysis of demand driven by corporate targets highlights a similar regional emphasis on vertically integrated areas as deadlines approach in later years (2030-2040).³⁷

4.3. Analysis by utility type

Virtually all city-level renewable energy demand before 2025 will come from cities with either retail choice, or their own municipal or cooperative utilities (Figure 7). This trend highlights the fact that cities with the greatest flexibility and decision-making authority in their renewable procurement options are the only ones setting commitments with deadlines at or before 2020. This is not to say that cities without retail choice are shying away from renewable commitments; their deadlines are just further out – allowing for time to work with (or in some cases, to find ways around) their vertically integrated electricity suppliers. In fact, come 2040, 73% (54TWh) of net demand will fall in cities served by industrial-owned utilities. While areas with retail choice account for 23TWh of this IOU demand, just under half of the municipal demand for renewables, (31TWh or 42%), will be in vertically integrated utility territories with no retail choice offering. In other words, a substantial amount of forecasted demand must be met through collaborating with (or in some cases around) industrial-owned utilities as cities falling in these areas do not have the ability to negotiate PPAs directly with other service providers.

Figure 7: U.S. cities’ annual net renewable electricity demand* by utility regulatory policy landscape

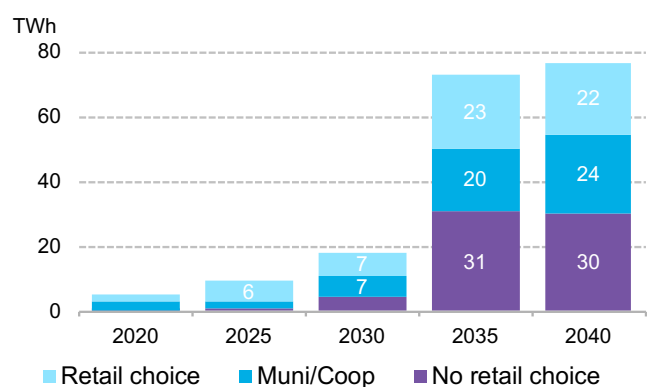


Figure 8: U.S. cities’ annual net renewable electricity demand by top five utility service providers

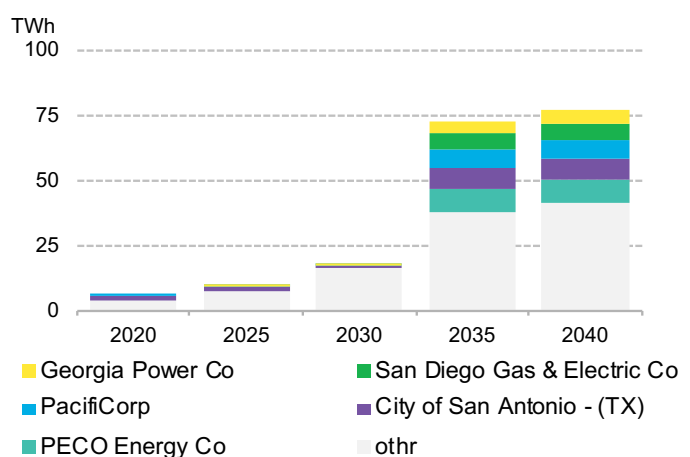


Figure 9 breaks the net renewable electricity demand from these 61 cities down by utility provider, showing the amount of demand that will fall on the top five load serving entities along with the aggregate amount across all other affected utilities. Four of the five entities that will see the most

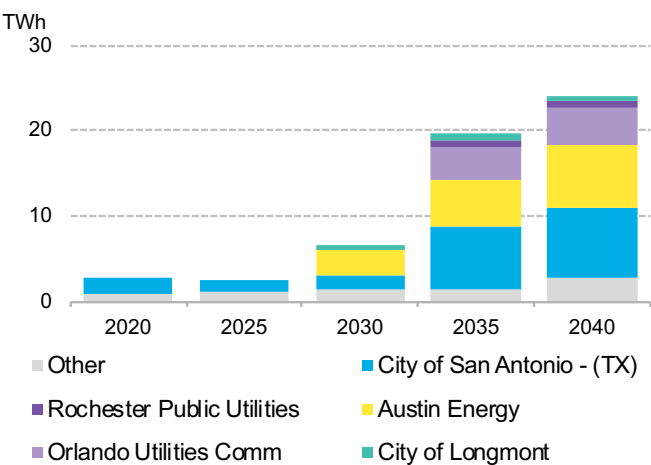
³⁷ Nicholas Albanese. (2016). What’s the growth opportunity for US corporate power purchase agreements? Bloomberg New Energy Finance. <https://www.bnef.com/core/insights/15295>.

increase in demand as a result of municipal renewable energy targets are industrial owned-utilities: PacifiCorp (serving Moab, Park City, Salt Lake City, and Summit, UT), San Diego Gas & Electric Co (serving Solana Beach, Chula Vista, and San Diego, CA), PECO Energy Co (serving Philadelphia, PA), and Georgia Power Co (serving Atlanta, GA).

Municipal, cooperative and rural utilities

Cities with municipal utilities have the most near-term demand for renewable energy, and it would be even higher had these utilities not already started contracting renewables. Their aggregate demand is proportionally higher relative to the other subset of cities, yet given the greater level of autonomy that municipalities have to procure renewables, these cities have already started procuring a good deal and thus the gap between their aggregate demand and the net “shortfall” of demand is narrowing at a faster pace than for other cities.³⁸

Figure 9: U.S. cities’ annual net renewable electricity demand by top five *Municipal utility* service providers



In 2035 three entities account for 85% of all municipal utility renewable energy demand: The City of San Antonio municipal utility with 7.3TWh, Austin Energy (5.2TWh), and the Orlando Utilities Commission (4TWh). Each of these three utilities have already started ramping up their renewables in response to consumer demands, with Austin Energy taking a significant lead bringing about 4GW of solar capacity online in 2015 alone³⁹. Table 4 below details a few examples of contracts incentivized by city energy commitments within municipal utilities.

³⁸ See the ‘Supply’ tab within the accompanying ‘US Cities RE Targets – Results and Underlying Analysis’ excel file 2 for a collection of the different types of contracts that have been signed by cities.

³⁹ Kyle Harrison. (2017). North America Clean Energy RFP Database. Bloomberg New Energy Finance. Bloomberg L.P. Terminal, New York, NY.

Table 4: Recent offsite power purchase agreements signed by municipal, cooperative and rural utilities to source renewables for city targets

Project	Utility	Capacity (MW)	Year signed	Year online	Tenor (years)
Midway Solar	Austin Energy - (TX)	170	2015	2018	25
West Wind	City of Denton - (TX)	100	–	2019	20
Pearl Solar	City of San Antonio - (TX)	50	2015	2017	25
Duke Orlando PV Plant	Orlando Utilities Comm - (FL)	5	2015	2016	20
Clear Spring Ranch	Colorado Springs Utilities - (CO)	10	–	2016	25
Borrego Easthampton PV Plant	Long Island Power Authority - (NY)	2.3	2010	2012	20

Note: assumes contract length to be 20 years unless otherwise indicated in project documents. All projects are solar except Denton city's West Wind contract. Source: See the 'Supply' tab within the accompanying 'US Cities RE Targets – Results and Underlying Analysis' excel file 2 for the raw data collected on different contracts and bibliographic links.

The majority of recent renewable supply brought online to comply with city energy targets has been a result of city owned utilities signing long term power purchase agreements on behalf of the city community as a whole – a unique ability only available to municipal utilities. Cities that don't own their own utility can still purchase renewable electricity for their public buildings directly through suppliers if they have retail choice, however they don't have the same ability as a municipal utility to do so on behalf of the citizens and customers within the city.

Section 5. Meeting the targets

There are many ways to procure renewable energy in compliance with set targets, and not all methods are equal. This section examines the way cities plan to meet their targets – whether it be through onsite generation, offsite or virtual power purchase agreements, or indirect methods such as purchasing RECs. This section also discusses historical trends in city-level renewable procurement, analysing the 18 cities that have previously achieved targets as potential indicators for the future.

5.1. Indicators from past achieved targets:

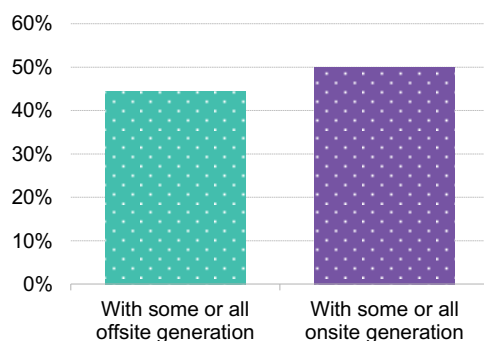
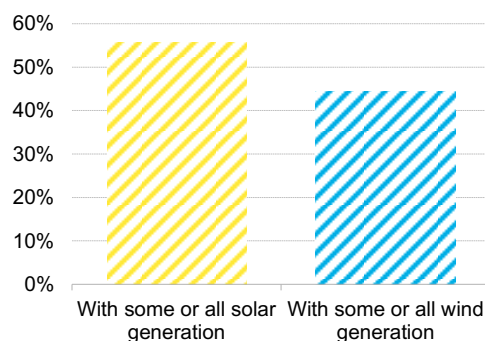
Within the 18 documented cities that have achieved their renewable energy targets, four (22%) did so with 100% RECs while 44% (8 cities) had a 'direct' approach, meaning that no voluntary RECs were counted towards achieving their targets. Instead, the cities used direct procurement of renewable electricity (either on or offsite) to meet their goals. Of these 8 cities with direct approaches, 3 achieved their targets with entirely onsite, and primarily wind, generation. Kodiak Island, AK, for example, worked with their cooperative utility to become 99.5% renewable in 2012 and has been supplying the entire community with 8GWh of onsite utility-scale wind and hydro ever since.⁴⁰

Table 5: How have 18 cities achieved their targets?

Approach	% of cities (out of 18)	Aggregate population	% of total population with achieved targets
Direct	44%	250,437	8%
Hybrid	33%	6,757,205	49%
RECs	22%	1,416,593	43%

Source: Interviews with city specific contacts and industry experts, publicly available data on individual municipal websites. See Appendix C.2 or the 'Achieved' tab within the accompanying 'U.S. City renewable energy targets and strategies dataset' for detailed data descriptions of each city's approach and city-specific sources. See Appendix B.1 for more details on sources and accuracy.

⁴⁰ EcoWatch. (2015). Second Largest Island in U.S. Goes 100% Renewable.
<https://www.ecowatch.com/second-largest-island-in-u-s-goes-100-renewable-1882043985.html>.

Figure 10: Percent of cities that achieved targets with on and/or offsite generation**Figure 11: Percent of cities that achieved targets with some or all wind or solar**

See the 'Achieved' tab within the accompanying 'U.S. City renewable energy targets and strategies dataset' for detailed data descriptions of each city's approaches, electricity profiles, and city-specific sources.

As demonstrated in Figures 11 and 12, in general, the majority of cities that have achieved their targets did so with relatively more onsite generation than offsite, and more solar (on or offsite) than wind generation.

While cities with direct approaches to meeting their targets make up the largest number (8 of the 18 achieved targets) these cities account for the smallest amount of population (only 8%) and none of them were in vertically integrated investor-owned utility areas. In other words, it may have been relatively easier for these cities to directly negotiate contracts for renewables than it will be for the cities with active targets that are served by IOUs in traditionally regulated markets, as municipal and rural electric utility structures offer more flexibility in initiating such contracts.

5.2. Meeting the net demand

If the past serves as any indicator for how cities will meet their targets moving forward, this data points to a heavy reliance on REC purchases. That being said, based on data collected from a series of interviews and city-specific public announcements, 29 of the 61 cities with active targets are currently planning to pursue direct approaches, and only 12 have indicated any intentions to use RECs towards their goals. While in theory purchasing renewable energy certificates can offset electricity use, REC markets in the U.S. are complex and ambiguity often surrounds the quality of certificates being purchased. Table 6 in the following section depicts a breakdown of the approaches that cities' have expressed intentions to pursue in meeting their targets.

For the purpose of simplicity, this analysis organizes intended approaches into three categories: 'direct', 'RECs' and 'hybrid'. "Direct" strategies involve a form of direct renewable electricity procurement – whether contracted through formal power purchase agreements (PPAs, either local or virtual) or onsite, through small-scale behind the meter generation. A 'RECs' approach implies that a city intends to attain their targets solely through the purchase of renewable energy credits. This method requires offsetting a city's electricity use by purchasing environmental attributes accredited to projects that reduce emissions through electric generation improvement. A

'hybrid' approach encompasses any combination of 'direct' renewable procurement and RECs or green-e certificate purchases.

A note on RECs and additionality: REC markets in the United States are complex and constantly evolving. There is a lack of transparency and often ambiguity surrounding the quality of the market. The degree to which a REC is 'unbundled' or 'bundled' with the source of its electric generation can affect the 'additionality', i.e., the creation of new renewable energy, of said metric.^{41, 42} REC prices also vary drastically across different electric regulatory zones, often depending on the level of demand for RECs, which can be driven by state-wide renewable mandates for utilities such as Renewable Portfolio Standards (RPS), as well as the abundance of supply. For instance, in ERCOT, the energy reliability council of Texas which is a market bubbling with solar, RECs can be as cheap as \$1.60 (compared to \$11.50 in PJM).⁴³ Subsequently, simply purchasing RECs can have varying degrees of impact in terms of emission reductions. For the purposes of this analysis, a 'RECs only' approach is treated as inferior to a more direct or hybrid approach, in terms of additionality—i.e., the creation of new demand for renewable energy that would not have otherwise existed.

How will cities meet their targets?

Cities are increasingly gravitating away from voluntary REC purchases and seeking renewable options that guarantee 'additionality'. RECs have played a large roll in past achieved city renewable energy targets to date – particularly within municipalities interested in offsetting electricity used for their government operations. For example, purchasing RECs to offset operations is currently all a place like Columbia, South Carolina has done in working towards their target.⁴⁴ However, it is unlikely that Columbia City is going to go out and purchase RECs to cover their entire community-wide electricity demand as there is no incentive or financial case to really do so. Based on stated targets and intentions expressed during interviews, this analysis does not expect RECs to play a huge roll in these community wide targets.

Intended method

In areas outside of the north-east (PJM, NYISO, and ISO-NE), cities have expressed intentions to meet their targets with direct strategies – signing long term PPAs, and pursuing onsite generation. While these strategies may change over time as deadlines approach, at the moment all cities whose primary electricity suppliers are either cooperative or community choice aggregators intend to meet their targets without the use of any REC purchases. 'Hybrid' approaches, incorporating both direct procurement and REC purchases, are more expected for cities served by IOUs or within more constraining regulatory landscapes.

⁴¹ City specific preferences for bundled and unbundled RECs were asked about during interviews and documented whenever information was available. See 'REC type' (column AN) in the Active, Looming, and Achieved tabs of the 'U.S. City renewable energy targets and strategies dataset' for data collected on this distinction.

⁴² Joshua Daniel. (2017). PJM Tier I REC Market Outlook. BNEF Bloomberg Finance LP, New York, NY. <https://www.bnef.com/core/insights/16899>.

⁴³ Nicholas Culver. (2013). The voluntary REC market after the flood. Bloomberg New Energy Finance. Bloomberg Finance LP, New York, NY.

⁴⁴ Columbia, SC Government Source. (2017). July 19th, 2017 Interview. Interviewer: Miela Mayer.

likely method

It is unclear if cities will stick to their intentions as deadlines approach. In the case of corporations making commitments to renewable energy, there are often public image considerations that factor into the decision of follow-through on commitments. For this reason, corporations struggling to meet targets once faced with the hurdles of direct renewable procurement will often still elect to follow through with such commitment by purchasing cheap, unbundled RECs.⁴⁵ However, it is arguable that cities and municipalities do not have the same 'green-washing' incentives. Thus, in the case that a deadline is approaching and a city has not achieved direct procurement, said city might forgo their commitment entirely before electing to simply purchase RECs as REC purchases do not offer much more than a stamp of green approval and could be hard to advocate for within municipal budget.

Table 6: Intended renewable electricity procurement strategies in cities with active targets, graphed by region and colored by electricity supplier type

	California	North-west	Mid-west	ERCOT	MISO	South-east	North-east
Direct	Lancaster_CA Los Angeles_CA Monterey_CA San Diego_CA San Francisco_CA San Jose_CA Santa Barbara_CA South Lake Tahoe_CA	Mercer Island_WA	Phoenix_AZ Boulder_CO Colorado Springs_CO Fort Collins_CO Albuquerque_NM County & Town of Taos_NM Moab_UT Park City_UT Salt Lake City_UT	Austin_TX San Antonio_TX	Columbia_MO Traverse City_MI Rochester_MN St Cloud_MN Madison_WI	Orange_FL Sarasota_FL Winter Park_FL Columbia_SC	Rockville_MD East Hampton_NY New York_NY
Hybrid		Portland_OR Multnomah_OR Edmonds_WA	Pueblo_CO	Denton_TX	Grand Rapids_MI	Orlando_FL St. Petersburg_FL Atlanta_GA	Chicago_IL Columbus_OH Philadelphia_PA Pittsburgh_PA Hanover_NH Ulster County_NY
	IOU or Retailer Municipal utility CCA or Coop						

Note: The above only displays cities' current intended paths towards meeting their targets, which may or may not align with the path they ultimately take. Only includes cities that have explicitly expressed intentions to pursue a certain approach, those with no preference stated for how they procure renewables have been excluded for accuracy purposes. Note also that a 'RECs only' approach category is not displayed along with the 'direct' and 'hybrid' grouping, as no city to date has announced (or expressed during interviews) intentions to only purchase RECs in order to comply with their energy targets.

To analyse more specifically how cities will actually procure renewable energy in compliance with their targets necessitates delving into the many factors at play – particularly the regulatory landscape. As eluded to in Section 3, electricity policies vary by location. In general cities with retail electricity choice, or that have their own municipal or rural cooperative utilities have more flexibility in renewable energy procurement. Beyond deregulation and retail choice, some states offer additional policies to facilitate renewable adoption. Such enabling policies include allowances for net energy metering (NEM), community-shared renewables, third party power

⁴⁵ Nicholas Albanese. (2016). What's the growth opportunity for US corporate power purchase agreements? Bloomberg New Energy Finance. <https://www.bnef.com/core/insights/15295>.

purchase agreements (PPAs), community choice aggregators (CCAs), and whether or not a state offers home rule authority.

NEM allows customers who own renewable assets on site (such as rooftop solar) to sell excess renewable generation back into the grid and be compensated either through a credit to, or reduction on, their monthly utility bill – thereby improving the financials for renewable investment. Virtual NEM, a certain subset of NEM policies, is of particular relevance to municipalities pursuing renewables. Virtual NEM policies (also referred to as aggregate NEM) enable customers with multiple meters, such as city-owned buildings, to be compensated for the generation at each meter.⁴⁶

In areas with virtual NEM allowances, communities can also form aggregate purchasing arrangements. These policies are thus particularly helpful for customers that may want to purchase renewable assets but do not have sufficient conditions on-site, as it enables selling generation back to the grid from off-site locations that may be more suited for renewables. Customers without space or the conditions for solar on their own rooftop can participate in community-shared renewables or community solar programs to purchase renewables through an aggregated-offsite arrangement when virtual net energy metering is allowed. Depending on the specific legislation, some areas may have system size limitations (meaning that one can only be compensated for up to a certain amount of generation).

Whether or not a city is located in a Home Rule or Dillon Rule state dictates the level of governing authority granted to municipalities. Cities in Home Rule states have more power over local decision-making (such as passing energy ordinances) than those in Dillon's rule states. In some cases in individual larger cities located within Dillon's Rule states can be granted Home Rule authority, as is the case for New York City.⁴⁷ Home rule cities might consider creating financial incentives for community investment in renewables, whether it's a tax break to those who own onsite generation or a rebate incentive program.

In vertically integrated states with no retail choice where customers don't have the ability to negotiate PPAs directly, these policies may be crucial in facilitating renewable adoption. Other enabling policies that may exist at the state level or be offered through the local investor-owned utility include: electric decoupling (designed to counteract utility disincentives for adoption of energy efficiency measures); direct access tariffs (enabling certain, usually large, customers to purchase renewables directly from retail suppliers despite being in a traditionally regulated market, the local IOU still provides transmission services); green power purchasing, and green pricing programs offered through the utility.⁴⁸

⁴⁶ Letha Tawney, Priya Barua, and Celina Bonugli. (2017). Emerging Green Tariffs in U.S. Regulated Electricity Markets. Washington, DC: World Resources Institute. http://www.wri.org/sites/default/files/emerging-green-tariffs-in-us-regulated-electricity-markets-sep2017_0.pdf

⁴⁷ Jon Russell and Aaron Bostrom. (2019). Federalism, Dillon Rule and Home Rule. ACCE (American City Country Exchange). <https://www.alec.org/app/uploads/2016/01/2016-ACCE-White-Paper-Dillon-House-Rule-Final.pdf>

⁴⁸ WRI. (2017). U.S. Green Tariff Deals (Maps and Data). <http://www.wri.org/resources/charts-graphs/grid-transformation-green-tariff-deals>.

Future outlook

5.3. Trends to look out for

This forecast of renewable demand, driven by city energy targets, assumes that current market conditions will prevail. The following section proposes potential market developments that may affect this forecast, and are key considerations for cities pursuing renewable electricity:

- **Evolving electric landscape:** The increasing shift towards electric vehicle fleets, particularly in cities that have set additional climate or carbon neutrality targets, could drastically increase the total electric loads that cities face, and thus increase their demand for renewable electricity.
- **Roll back on federal tax credits for renewables:** The Investment Tax Credit (ITC) and the Production Tax Credit (PTC) are scheduled to phase out in 2021 and 2020 respectively, leading to a potential spike in wind and solar costs. With these pending expirations in mind, utilities may be more eager to sign long term contracts for renewables in the coming years. There is a strong incentive for cities to frontload their contracting and meet targets early while the federal tax credit is still available, in order to lock in favourable costs.
- **Electricity price variability:** Electricity prices are important drivers for wind and solar development. Therefore, falling prices across the country could render wind and solar developments less economically competitive.
- **Community choice aggregation:** Community choice aggregation leverages collective buying power of multiple electricity users to set up a separate entity that is allowed to sell electricity in the state – this electricity is usually able to be cleaner than the standard provided by the existing local utility. CCAs have mostly dominated the market in California and proven to be quite successful in providing renewables for places like Monterey and Santa Cruse, CA.⁴⁹ However, recent legislation has been passed making CCAs allowed in 6 other states outside of California, with similar laws underway for Utah and Minnesota. CCAs could thus play a growing role in cities meeting their renewable targets if the feasibility of such community utility structures becomes further flushed out within these other states.
- **Trade tariff on solar:** In January of 2018 the federal government determined that cheap, imported solar panels were harming domestic production enough to warrant a 30% trade tariff on all imports of solar cells and modules.⁵⁰ Effective as of February 2018, this tariff is expected to last 4 years and decrease by 5% each year to a final 15% in 2021. According to the U.S. Trade Representative fact sheet, the first 2.5GW of cell imports in each fiscal year will be exempt from the tariff.⁵¹ In contrast to the roll back of the federal tax credits for renewables, this tariff could incentivize delaying (rather than frontloading) solar deals until

⁴⁹ City of Solana Beach, California. (2017). Community Choice Aggregation (CCA). <http://www.ci.solana-beach.ca.us/cca>. [Accessed December, 2017]

⁵⁰ Julia Pyper. (Jan 2018). Trump Administration Issues 30% Solar Panel Import Tariff. *Green Tech Media*. <https://www.greentechmedia.com/articles/read/breaking-trump-admin-issues-a-30-solar-tariff#gs.XO9tmebR>.

⁵¹ U.S. Trade Representative Robert Lighthizer. (2018). Section 201 Cases: Imported Large Residential Washing Machines and Imported Solar Cells and Modules. <https://ustr.gov/sites/default/files/files/Press/fs/201%20Cases%20Fact%20Sheet.pdf>.

after tariff has decreased. This is because the tariff harms the economics of utility scale solar contracts.

Taking the above factors and market trends into consideration, there is still hope for rhetoric to become reality. Maximizing adoption to meet these commitments is not impossible, but it will take dedicated and unwavering efforts to stay the course given the long-time horizon. Progress can still be achieved through collaboration and mutual good will between municipalities and their utilities.

Austin provides a compelling example of successful renewable energy adoption. By cultivating shared values and a shared desire for cost-effectiveness, Austin has procured significant renewables through their municipal utility, Austin Energy. However, Austin has benefitted from significant advantages, such as a favourable regulatory landscape, and a local abundance of solar energy. However, while Austin provides an encouraging example, not all cities are as well positioned to meet their renewable demand.

While the most effective renewable adoption would certainly be facilitated if all cities were able to be like Austin energy – which has procured significant renewables through their municipal utility due to aligning incentives both in terms of values and cost effectiveness – not all cities are fortunate enough to have such ample renewable conditions and enabling regulatory structures.

Yet despite the absence of tax incentives, and the highly variable institutional and economic structures, all is not lost. There is still a good deal of opportunity and hope for cities facing even the most constraining regulatory conditions. As discussed in the previous section, there are still many enabling policies available to cities with IOUs and no retail choice. Beyond such policies, there are also inspiring examples of cities going above and beyond to reach their goals, even in the most unfavourable regulatory landscapes.

For example, despite being serviced by a traditional coal utility, Park City and Salt Lake City have joined together to successfully collaborate with their IOU, Rocky Mountain Power (RMP), in procuring renewables. While RMP had no obligation to engage in providing renewable electricity to these cities, they were able to work collaboratively to convince RMP to sign a Joint Clean Energy Cooperative Agreement.⁵² This is a particularly hopeful example given that Park City is simply too small to qualify as a key utility account giving it less bargaining power than larger cities like Boston or even Yale.⁵³ Nonetheless, both Park City and Salt Lake City were able to work together with RMP in good faith to reach a mutually agreeable outcome.⁵⁴ Even more impressively, these cities are not only setting targets for renewable energy, but pledging to become net-zero by the same deadline. In their commitment to do so, they define 100% renewable on annualized basis as entirely new-build. This means that reaching their target excludes any existing or planned renewable energy supplied through RMP to date, as well as existing unbundled RECs. In other words, cities are voluntarily raising the bar, striving to become not only carbon neutral, but carbon negative – not even counting existing supply in the mix. This case study demonstrates the power of cooperation, both with neighbouring cities and with local utility companies. The striking, and perhaps even unique, level of volunteerism that characterizes

⁵² Salt Lake City. (2017). RMP Implementation Plan. <http://www.slccgreen.com/rmp-implementation-plan>

⁵³ Nicholas Albanese. (2016). US utilities key accounts database. Bloomberg New Energy Finance. Bloomberg Finance L.P., New York, NY.

⁵⁴ Currently no paths exist for CCA in Utah but joint agreement commits RMP to providing similar options under the premise that the city does not seek programs elsewhere.

the carbon-neutral movement is one reason to be hopeful that cities may yet have some success in meeting their projected demand.

The city of Sarasota, FL, provides another example of the creativity and resourcefulness that some cities have exhibited in the face of unfavourable regulatory conditions. Sarasota purposefully set its deadline to be 5 years after the renewal date for their 30-year franchise agreement with their local IOU, Florida Power and Light (FPL). While Sarasota hopes to work with FPL in procuring renewables for the community-wide target, the city is also open to renegotiating or even not renewing their franchise agreement with FPL, should that be necessary. The city has considered municipalizing its electric utility in the past, and this may be feasible because it is located in a home rule state.^{55, 56}

Even for many of the cities discussed here that are working within the most constrained regulatory landscapes, i.e., those in vertically integrated industrial-owned utility service territories, there is hope for effective renewable adoption. See below for a list of some of the more creative strategies available to cities in restrictive regulatory landscapes, and examples of cities pursuing these strategies.

Possible options available to cities in restrictive regulatory landscapes:

- **Leverage purchasing power to negotiate renewable options with the local utility**
 - **Leveraging purchasing power:** Some cities within the subset analysed are large enough to qualify as key account holders to their respective utilities. In these cases the cities are in a unique position to leverage their relatively large purchasing power to negotiate agreements with their utilities. Of the 61 cities in this dataset, many are key utility customers within their service zones, and thus may be able to negotiate a better renewable mix in the portfolio of energy supplied.
 - *Example:* the city of Monterey, CA is key customer of Commerce Energy, Inc. Additionally, Boston, MA, a city which already achieved a municipal level target,⁵⁷ is a key customer of Nstar Electric Co. If the city were to consider pursuing a community-wide commitment, they could utilize their position as one of their utility's main customers to achieve such a goal.⁵⁸
 - **Combining forces:** some cities that may not be large enough to influence their utilities alone have had success joining with neighbouring communities to do so.
 - *Example:* Park City and Salt Lake City, UT joined together to work with RMP as discussed above
 - This tactic is not just relevant to cities and towns, but rather any entities interested in influencing their electricity mix. For example, Rockville, MD might consider combining forces with the University of Georgetown as both the city and university have indicated intentions to increase their renewable energy, and both purchase their electricity from Potomac Electric Power Co. Georgetown is large enough to be listed as a key utility account and thus such a partnership might help Rockville to achieve their own goal. The

⁵⁵ Sarasota, FL Government Source. (2017). July 18th, 2017 Interview. Interviewer: Miela Mayer.

⁵⁶ While Florida has home rule state legislation, this legislation excludes taxing authorities.

⁵⁷ See the achieved tab in excel file #1 accompanying this research for more details on the 18 documented cities to have met their goals.

⁵⁸ Federal Energy Regulatory Commission (FERC). (2016). Form 566: Annual Report of a Utility's 20 Largest Customers. <http://www.ferc.gov/docs-filing/forms.asp#566>.

same would be true for cities in Massachusetts if they were to collaborate with Williams College to work with NextEra energy.⁵⁹

- **Renegotiate franchise agreements with local utility service providers**
 - *Example:* The county and town of Taos, NM are seeking to exit their agreement with the Tri-State Generation and Transmission Association so that they may source renewables through the Kit Karson Electric Cooperative (KCEC). This is a joint effort between multiple cities and towns in the KCEC service area, led by Renewable Taos, Inc.⁶⁰ Their strategy is to leverage a June 18, 2015 Federal Energy Regulatory Commission ruling in order to terminate the contract, which currently requires KCEC buy 95% of its energy from Tri-State Generation and Transmission Association through 2040.⁶¹
 - *Example:* As discussed above, Sarasota's 30-year franchise agreement with FPL is up for renewal in 2040, 5 years before the city's renewable target deadline. The city has openly considered exiting this franchise agreement in order to meet their target if such measures become necessary.⁶²
- **Municipalize the local utility:** As a last resort, cities may pursue municipalization (i.e., creating their own municipal utility despite the local regulatory landscape). This process involves the city government purchasing the existing private utility to create a municipally-owned entity.
 - *Example:* Boulder, CO has been working to separate from their investor-owned utility, the Public Service Co of Colorado (PSCo) since 2010. In 2016 the city filed applications to the Colorado Public Utilities Commission to establish a municipal electric utility and pass renewable energy offset requirements.⁶³
- **Create a community choice aggregator as an alternative to the local utility**
 - *Example:* CCAs have proven to be incredibly successful in providing greater portions of renewables to many cities in California.⁶⁴ 19 cities with unmet demand are in areas with CCAs are allowed - 6 of these cities are located outside of California. This tool could become options for cities in New York (NYC, Ulster County and East Hampton), Ohio (Cincinnati and Columbus, OH) and Chicago, IL. An additional 6 cities fall in areas in the process of passing CCA laws.

⁵⁹ Nicholas Albanese. (2016). US utilities key accounts database. Bloomberg New Energy Finance. Bloomberg Finance L.P., New York, NY.

⁶⁰ Additional towns and villages part of this joint effort include: Angle Fire, Eagle Nest, Questa, and Red River.

⁶¹ Renewable Taos. (2014). Joint Resolution on Renewable Energy. Accessed 2017. <http://renewabletaos.org/sites/default/files/field/pdf>.

⁶² Sarasota, FL Government Source. (2017). July 18th, 2017 Interview. Interviewer: Miela Mayer.

⁶³ City of Boulder, Colorado. (2017). Boulder, Xcel Energy Acknowledge Settlement Discussions. <https://bouldercolorado.gov/newsroom/june-8-2016-boulder-xcel-energy-acknowledge-settlement-discussions>

⁶⁴ City of Solana Beach, California. (2017). Community Choice Aggregation (CCA). <http://www.ci.solana-beach.ca.us/cca>. [Accessed December, 2017]

5.4. Can rhetoric become reality?

Do these targets indicate new demand for renewable developers, a loss of customers for traditional electric utility providers, or are they just hot air?

The impact of city-level renewable energy targets ultimately depends on the level of initiative taken to meeting them. 74TWh is not insignificant – assuming 40% of this figure is met with wind and 60% is met with solar, the resulting demand for renewables would require 27GW of new capacity to be online by 2035. Such an increase in renewable energy production would undoubtedly be a substantial step in the right direction.

That being said, no commitment is legally binding. While a handful of cities have published formal sustainability reports or energy action plans laying out their intentions to make good on their targets, the vast majority have not. Most have not moved far beyond voicing a target, and in even the best cases, setting a firm deadline. Achievement rates will become more apparent as deadlines approach, but ultimately even the most formal commitments are relatively toothless, and do not specify consequences for unmet or delayed commitments. It is therefore always possible that some cities fall well short of their targets, especially given the long-time horizon of their efforts.

Yet there may still be reason to hope. Indeed, city behaviours have been anything but predictable. Contrary to what one might expect, cities who face favourable conditions are not the only municipalities announcing targets. Surprisingly, even cities such as Atlanta, GA have made renewable commitments.⁶⁵ Atlanta is an unlikely candidate for such a commitment because it sits within a vertically integrated IOU territory, and is located in state with few policies that encourage the adoption of renewables. Yet such cities are making ambitious commitments, despite there being no clear path to renewable procurement or precedent for achieving their targets. This may seem disheartening, but more and more success stories are emerging from such unlikely scenarios, especially as the need to mitigate climate change is felt more urgently within these communities. The fact that a broad range of cities are entering into commitments across the country is a rare and encouraging indicator of widespread willingness to engage in climate change and to work for a better future.

The timeline of announcements presented in the first section, and the uptick in city commitments in 2016 underscores the momentum that this popular movement to save the planet has gained. There is clearly a lot of enthusiasm for cities entering into these commitments, if for no other reason, to signal the underlying values of such a goal. In the absence of legal coercion or clear economic incentives these cities behaviours might be thought of as exemplary of altruism or as simply motivated by the imperative to do the right thing.

Even if some renewable commitments simply function as a manifesto of what is wanted, and less on what is immediately feasible, these commitments may still have a positive impact. Regardless of the motives driving each individual city—whether it be the imperative to do the right thing, a new sense of urgency about reducing environmental impact, or the desire to signal support for a popular movement to save the planet – the result is a significant new source of demand for renewables. And the momentum is only growing: more and more cities in unconventional situations are proving that it is possible to set a goal and find solutions despite their varied institutional, financial, political and economic constraints.

⁶⁵ EcoWatch. (2015). Second Largest Island in U.S. Goes 100% Renewable.
<https://www.ecowatch.com/second-largest-island-in-u-s-goes-100-renewable-1882043985.html>.

Indeed, even the most toothless commitments may still result in positive change. This is supported by the fact that there is a large and growing number of cities that are procuring renewables without having made any official commitment to do so. For example, despite not having an official target, Los Angeles has made substantial strides in adopting renewables. The city's municipal utility LADWP has purchased an impressive quantity of wind and solar over the past few years (totaling approximately 2,550 MW).⁶⁶ A list of just a few of these many cities can be found in the 'looming' tab of the excel file #1 accompanying this analysis. The fact that some cities are making impressive strides regardless of any formal commitment to do so is one reason for hope and highlights what an exemplary demonstration of volunteerism is growing out of this movement. It indicates that some cities may rise to the occasion and may in fact meet their targets despite the absence of any substantial consequences if they do not. The behavior of cities like Los Angeles suggests that the looming threat of climate change may be consequence enough to galvanize action.

This trend of setting renewable energy targets goes far beyond cities (and corporates). Other entities to watch out for include universities – particularly those that are key customers for their respective utilities such as Williams college – states, and school districts. Hawaii as a state has already committed to be 100% renewable⁶⁷ and governor Jerry Brown is considering implementing a 100% renewable RPS (mandatory) for the state of California.⁶⁸ There have even been some city councils in North Carolina committing their entire state to renewable energy. While a city's authority to see such a claim through is ambiguous, the intentionality and urgency of these statements are impactful on their own. If such trends are to continue we could have much more demand. Utilizing an appropriate mix of the best practices identified in this study will increase the likelihood for cities to successfully set and meet their renewable energy goals.

⁶⁶ LADWP. (2018). Renewable Energy Program. https://www.ladwp.com/ladwp/faces/ladwp/aboutus/a-power/a-p-renewableenergy/a-p-renewableenergypolicy?_afLoop=366184256590619&_afWindowMode=0&_afWindowId=null#%40%3F_afWindowId%3Dnull%26_afLoop%3D366184256590619%26_afWindowMode%3D0%26_adf.ctrl-state%3D1ir9qiyar_17

⁶⁷ Julia Pyper. (May 2015). State Bulletin: Hawaii Passes Legislation to go 100% Renewable. *Green Tech Media*. <https://www.greentechmedia.com/squared/state-bulletin/hawaii-passes-legislation-to-go-100-renewable1>.

⁶⁸ Meredith Rutland Bauer. (2017). California to pass law setting 100% renewable electricity goal. *Climate Change News*. <http://www.climatechangenews.com/2017/08/31/california-pass-law-100-renewable-electricity-goal/>

Appendices

Appendix A. Methodology and assumptions

The purpose of this analysis is to define the universe of formally adopted commitments to renewable energy by U.S. cities and to forecast the minimum amount of new demand driven by these city targets through 2035. Please see the accompanying 'U.S. City Renewable Energy Targets and Strategies Dataset' for raw data collection and the 'U.S. Cities RE Targets – Results and underlying analysis' Excel file for a direct look at the analysis and demand forecast for individual cities. For further details on how to navigate these Excel files and the demand visualization tools incorporated, see Appendix B.

A.1. City selection

This analysis examines a diverse selection of 61 U.S. cities with formally adopted commitments to a substantial amount of renewable energy. This subset of 61 'active' cities – ie, cities actively pursuing renewable energy targets with deadlines in the foreseeable future – was identified using the criteria outlined below.

Selection criteria – 'active' cities

Each qualifying U.S. city:

- Has a population equal to or greater than 5,000 residents
- Has formally adopted a renewable energy target that accounts for at least either 10% of community-wide electricity demand or 20% of municipal operations electricity demand

This study selected the above criteria in order to encompass a selection representative of the cities most likely to drive renewable energy opportunities through their newly targeted demand. Cities that are actively procuring substantial amounts of renewable energy despite not having formally adopted any explicit targets or deadlines are also tracked in the data collection of this research as 'looming' cities but are excluded from the demand forecast for active city targets. For more information on actions being taken by 'looming' cities as well as the 18 cities that have achieved their energy targets, see Appendix C for the accompanying 'U.S. City Renewable Energy Targets and Strategies Dataset' detailed in Appendix B.1.

A.2. Estimating the U.S. cities' incremental demand for renewable energy

The goal of this research is to forecast the minimum amount of new demand driven by city renewable energy targets through 2035. The analysis examines stated renewable electricity goals of 61 'active' cities and their respective electricity demand in order to estimate the renewable electricity required for compliance in each year through 2035. The forecast method consists of three stages with subsidiary steps detailed below.

Key assumptions

- Entities with populations below 5,000 are excluded from the analysis of active city targets for the purposes of scale⁶⁹.
- All targets are achieved by 2035 at the latest. This assumption applies to any cities with undefined deadlines or deadlines set beyond 2035 (7 instances in total, as noted with an asterisk (*) in Table 11 within Appendix C).
- Progress towards targets is made on an as-needed basis, with no linear ramp up imposed aside from that induced by stated interim targets⁷⁰. Each target is met on an annual basis after the initial deadline year through 2035 – and presumably thereafter.
- Annual electricity demand growth for each city through 2035 equals the 2010-2015 electricity sales CAGR of their respective primary utility provider.
- Contracted renewable generation stays online through the contracts' tenor – assumed to be 20 years in the absence of data– and contracts are not renewed or further extended after their expiration. Onsite generation remains flat year-over-year.
- The percentage of a cities' electricity load that is offset with voluntary renewable energy credit (REC) purchases is consistent with their historic purchasing trends and remains constant through 2035.
- All cities count any renewable energy that is sourced for compliance with state renewable portfolio standards (RPS) towards their renewable electricity targets in states with such standard. For consistency and providing a conservative estimate, this assumption applies even to cities that have indicated otherwise for 'additionality' purposes (i.e., to drive purely new renewable generation).
- The electricity demand from a city's municipal operations is equal to 3% of the city's total electricity demand⁷¹.

Assessing aggregate demand (stage 1 of 3)

- *Step 1:* identify the relevant electricity load targeted by each city's commitments (municipal or city-wide) based on stated and estimated values.
 - Stated values come from publicly-available local government websites, sustainability or climate action reports, and announcements. Extensive interviews were conducted to improve the accuracy of the model's electricity demand figures (and to gather information on each cities' approach to meeting their targets) whenever possible⁷².
 - For cities with their own municipal utility, the model utilizes the utilities' 2015 electricity sales, collected from EIA's [Form 861](#), as the city's total electricity demand.

⁶⁹ Populations below the threshold of 5,000 occupants are excluded for the purposes of scale. Towns and villages excluded due to small population size but that have formally adopted renewable energy targets include: Del Mar, CA; Nederland, CO; Abita Springs, LA; Nassau, NY; Southampton, NY; Castle Valley, UT; and the New Mexico villages of Angle Fire, Eagle Nest and Red River.

⁷⁰ 18 cities have set one or more interim commitments.

⁷¹ 'Municipal operations' refers to the electricity demand from city owned buildings and facilities, rather than the entire city's electricity demand – see the US Cities RE Targets and Strategies database for further details on this distinction.

⁷² See the raw data collected within the accompanying U.S. City Renewable Energy Targets and Strategies dataset for additional details on individual city approaches and documented actions to-date.

- In the absence of post-2015 reported values, the model estimates electricity demand by multiplying each cities' 2016 population by the average 2015 electricity sales per capita in their respective states. Population figures come from the [U.S. Census Bureau](#)⁷³ and the model's state sales per capita figures are based on annual [EIA data](#).
- *Step 2: forecast the incremental aggregate demand for renewable electricity incentivized by U.S. city targets through 2035 by:*
 - Calculating the 2010-2015 electricity sales CAGR of each city's primary electric utility provider to forecast the total electricity demand in each year from 2015 to 2035⁷⁴.
 - Identifying, based on stated targets and deadlines, the percentage of aggregate demand that each city has committed to supply with renewable electricity.
 - Converting this percentage into annual aggregate demand for renewable electricity by applying the target percentages to each cities' total demand year-over-year through 2035, starting with their target's initial deadline year (or interim percentage and deadline where available).

Assessing the portion of aggregate demand already accounted for (stage 2 of 3):

To detail what is effectively the supply side of the story the model sums the amount of demand covered by existing or contracted generation, the amount likely to be met by REC purchases, and the portion already spoken for by state level RPS.

Step 1: forecast existing and contracted renewable generation by:

- Reviewing local government publications, as well as municipal documents via the Bloomberg Terminal, and Bloomberg New Energy Finance's [US corporate PPA project database](#), [California RPS contracts database](#), and [North America Clean Energy RFP Database](#) to identify existing contracts or purchases on the books for renewable generation.
- Determining the capacity, start date, and tenor of each contract and applying actual and estimated technology-specific capacity factors to convert from contracted capacity to estimated onsite and offsite annual generation in each year through 2040⁷⁵.
- *Note:* see the 'Supply' tab in the 'U.S. Cities RE Targets – Results and underlying analysis' Excel file for the PPAs and other renewable contracts included in this analysis.

Step 2: estimate the amount of demand likely to be met with REC purchases by:

- Determining the amount of RECs each city has historically purchased based on stated values and information gathered from the EPA's [Green Power Partnership](#) database.
- Converting this amount to a percentage of 2015 electricity demand and applying that percentage to the demand in each year, under the assumption that purchasing trends remain proportionally constant.

Step 3: determine the percentage of demand already spoken for by state RPS by:

⁷³ In cases where a both a city and the county in which it is located (or vice versa) are pursuing renewable energy commitments—such as in Portland and Multnomah, OR—the model subtracts the city's population from that of the county to avoid double counting.

⁷⁴ Electricity sales data for each utility comes from EIA [Form 861](#). In the event that annual electricity demand for the city itself was available for these years, the model preferences the cities' demand CAGR over that calculated from electricity sales.

⁷⁵ All estimated values are based on the U.S. EIA 2016 annual averages: 35% Wind, 27% Solar PV, 38% Hydroelectric, 71% Landfill Gas and Municipal Solid Waste, 47% Biomass, and 75% Geothermal. Actual capacity factors are based on generation disclosed in official documentation.

- Identifying which cities fall in states subject to RPS and assuming that all cities will count renewable energy sourced for compliance with the standards towards their renewable electricity targets.
- Applying the relevant Tier 1 RPS percentage to each cities' total electricity demand, not just those served by investor-owned utilities, to determine the annual demand effectively met with renewable sources.
- Note that this study only account for the mandatory Tier 1 RPS percentages – which generally count RECs generated from new wind, solar and biomass sources towards compliance. The study chose not to include Tier II RPS—i.e., generation from older facilities or existing hydro—or any voluntary standards. RPS target percentages and incremental schedules for each state are consistent with Bloomberg New Energy Finance's US renewable portfolio standard (RPS) demand database.

Assessing the net demand incentivised by city renewable energy targets (stage 3 of 3):

To arrive at the 'net renewable energy demand' – i.e., the portion of this aggregate demand that may have implications for, say, a developer – the model calculates the difference between U.S. city renewable electricity targets and the total renewable supply (REC purchases, existing or contracted onsite and offsite generation, and renewables from RPS) over 2015-2035 for the 61 cities.

Appendix B. Accompanying Excel Documents

Two excel documents accompany this research – the first is a descriptive dataset containing collected information and citations for each city's target, overall approach, electricity profile, and relevant regulatory information. The second document is the demand analysis upon which all market sizing conclusions are based.

B.1. Excel file 1: Dataset of collected information on cities' renewable energy targets, strategies, electricity profiles, and regulatory information

This Dataset captures renewable energy targets and commitments made by 79 cities, counties, towns and other municipal entities across the United States. It details U.S. cities and municipalities' renewable procurement strategies and provides a rich set of information and data relevant to understanding cities' current and future energy demand profiles.

What is captured

The accompanying Excel spreadsheet includes known U.S. city and municipal-level renewable energy targets, highlighting electricity supply information and relevant regulatory structures.

Two types of city renewable energy targets

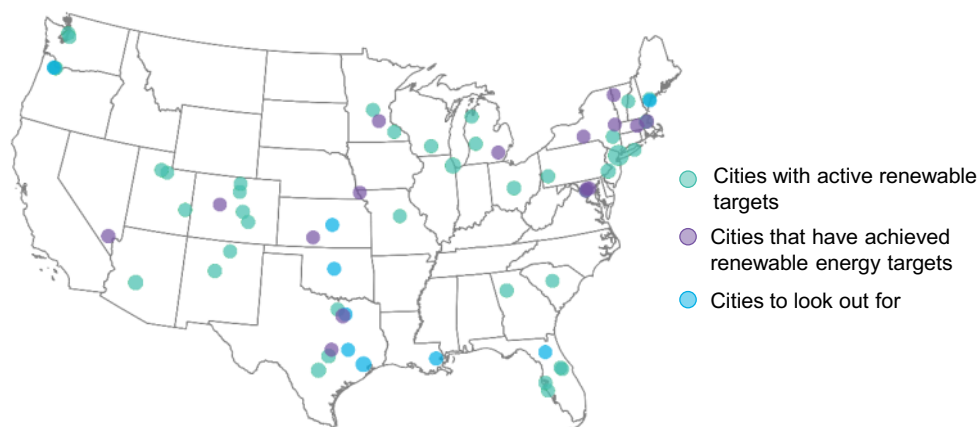
City renewable energy targets generally fall into one of two categories:

3. Commitments for *municipal operations*, targeting renewable energy to supply the electric load of city government buildings and facilities;
4. Commitments for *community-wide operations*, which encapsulate electricity demand not just for the city's own operations but for the citizens and businesses within the city as well.

In some cases, cities commit first to a target covering their municipal operations as an interim step toward a target covering their community-wide operations. The accompanying dataset breaks out municipal operations targets separately from community-wide operations targets to highlight this difference. Table 8 below outlines how this distinction is handled in the dataset.

A note on terminology

The term 'city' is used throughout this document and in the corresponding spreadsheet to refer to the various entity types covered by this analysis: cities, towns, counties, villages, boroughs, and municipalities.

Figure 12: Cities included in this dataset colored by renewable energy target status

Source: Map generated in QGIS based on data collected from various city and climate action public announcements, interviews, and the Bloomberg Terminal. Notes: Categories described in Table 7 below. Excludes all cities in California and small populations (<5,000 residents).

Not captured: California Cities and Community Choice Aggregators (CCAs)

While many California cities are included in the demand analysis of this study, the descriptive dataset currently excludes all cities with renewable energy targets in the State of California, where most are pursuing community choice aggregation (CCA). This exclusion is due to the complexity and rapidly evolving nature of the new CCA landscape.

Sources and accuracy

The information presented in this dataset is subject to known limitations, the most prevalent being variability in the quantity and quality of up-to-date entity-specific data. The majority of data is collected from publicly-available local government websites, publications, and announcements. Electricity supplier information is consistent with annual data released by the Energy Information Administration (EIA), specifically [Form 861](#).

Other sources include – but are not limited to – the following: the Database of State Incentives for Renewables & Efficiency (DSIRE), the Federal Energy Regulatory Commission (FERC), the National Renewable Energy Laboratory (NREL), the Environmental Protection Agency (EPA), the Environmental Defence Fund (EDF), the Sierra Club, and the Carbon Neutral Cities Alliance (CNCA). Field-specific sources are noted in the following explanatory tables.

Field descriptions and abbreviations

This dataset breaks the 79 captured city and municipal-level renewable energy targets out into three tabs in order to separate cities actively pursuing targets ('Active') from those that have achieved their renewable energy targets ('Achieved') and fourteen cities that this study believes have future demand for renewable energy despite lacking formally adopted commitments ('Looming'). Table 7 outlines each of the three tabs in the accompanying Excel.

Table 7: Excel file #1 – Description of tabs

Tab (excel sheet)	Explanation
Active	Captures 49 cities that have made renewable commitments with deadlines in the foreseeable future. Cities with populations below 5,000 are excluded from this section for purposes of scale.*
Achieved	Captures 18 cities that have achieved their renewable energy targets. A population minimum is not imposed in this list so as to encompass the range of strategies employed.
Looming	Includes 14 progressive cities that, despite not having formally adopted any explicit targets, are taking concrete actions to increase their renewable procurement (often as part of broader sustainability or climate action plans).

Note: () Towns and villages excluded due to population size but that have formally adopted renewable energy targets include: Nederland, CO; Abita Springs, LA; Nassau, NY; Southampton, NY; Castle Valley, UT; and the New Mexico villages of Angel Fire, Eagle Nest, Questa, and Red River.*

Table 8: Excel file #1 – Description of fields in each tab

Column	Field Title	Explanation
C	Entity	Name of the relevant city, county, town, borough, or village
D	Entity type	Dataset is primarily comprised of city-level targets, but some counties, towns, and boroughs are also included. All entities with populations below 5,000 are excluded from the list of active targets.
E ¹	State	State in which entity is located. Currently excludes cities with renewable energy targets in the State of California, where most are pursuing community choice aggregation.
F	Electricity supplier type	Categorization of the relevant electricity supplier as a municipal utility (Muni), investor-owned utility (IOU), electric utility cooperative (Coop), retail electric supplier (Retailer), or political subdivision. For cities in which electricity choice is available this field may only list the top suppliers.
G	Electricity supplier	Name of the relevant electric utility. Parent companies may be stated in the absence of subsidiary data. Official public titles are used whenever possible, and thus there may be inconsistencies with the EIA's notation.
H ²	Electricity supplier EIA ID	Form 861 identification number as used by EIA.
I	Estimated electricity demand	Relevant municipal or community-wide electricity load targeted by the entities' commitments. In areas served by a municipal utility, demand is consistent with the utilities' 2015 electricity sales, collected from EIA's Form 861 . In the absence of a city-owned utility or city-specific post-2015 reported values, electricity demand figures are estimated based on each cities' 2016 population (U.S. Census Bureau) and the average 2015 electricity sales per capita in their respective states (EIA data).
J	Electricity demand scope	Denotes whether the electricity demand figure in Column J pertains to municipal operations only (Municipal-only) such as city government owned buildings and facilities, or if it is a full city figure (Community-wide) encompassing the entire load of city residents and businesses.
L	Projected RE demand	Rough estimate of the renewable demand incentivized by targets—figure is based purely on target (% or MWh) and the most recently documented existing demand.
M	Projection year	Correlated directly to the relevant commitment deadline year, assumes flat demand over time.
O	Community-wide renewable target	The target percent of renewables to be incorporated into the electricity/energy supply for the entire community's operations (residential, commercial, industrial, municipal, etc.) based on public commitments.
W	Municipal renewable target	The target percent of renewables to be incorporated into the electricity/energy supply for a city's municipal operations only (city owned buildings, government facilities, etc.) based on public commitments.
P, T, X, AB	Target unit	Denotes whether target percentages apply to the city's electricity demand (% Electricity) or energy use as a whole (% Energy). Simply a percent symbol (%) signifies that the scope is ambiguous and

Column	Field Title	Explanation
		is assumed to pertain to electricity demand for the purpose of future analysis. In the cases where commitments are made in alternative units with limited conversion data, the original unit is provided.
Q, Y	Announced	Announcement date of each commitment. Assumed to be the 31st of December in cases where only the year is available.
R, Z	Deadline	Calendar year by which each goal is set to be achieved.
AE ³	Status	Categorization of cities' progress towards meeting their targets into: • Planning – in the early stages with little or no action towards target documented to-date • In progress – concrete steps or initiatives are underway towards procuring renewables in compliance with the target • Achieved – city has met their renewable energy target
AF	Approach	Categorization of cities' approaches to meeting their targets into: • Direct – Signs long-term PPAs (onsite or offsite) or owns systems directly • RECs – Purchases voluntary RECs to achieve stated renewable goals • Hybrid – Combination of 'direct' procurement and REC purchases • TBD – To be determined (TBD/Direct or TBD/Hybrid simply indicates that the city's historic trends may lend more readily to one strategy or the other, but the entry is only speculative)
AI	Existing RE (%)	Represents the current percentage of renewable energy or electricity procured by each city through PPAs or onsite generation (based on most recently available data year); excludes REC purchases.
AJ	Scope of existing RE	Determines whether the current percentage of renewable energy documented is a percentage of municipal operations (Municipal-only) or a percentage of all city operations (Community-wide).
AK	Existing RE (GWh)	Represents the existing amount of renewable energy or electricity procured by each city through PPAs or onsite generation (based on most recently available data year); excludes REC purchases.
AM	REC purchases (MWh)	Quantifies in megawatt-hours (MWh) the Renewable Energy Credit purchases made by each entity. Figures encompass REC purchases, green power marketing, and green pricing programs as reported during interviews or documented by the EPA Green Power Purchasing Program and NREL.
AN ⁴	REC type	Denotes whether RECs purchased come alongside clean energy in their contracts (bundled) or if the city is willing to procure unbundled RECs. • Bundled • Unbundled • Blank or TBD – no expressed preference known to date
AP	2016 population	All population figures are sourced from the U.S. Census Bureau's 2016 projections, based on data collected in 2010.
AQ	ISO/RTO	Denotes (where available) the relevant independent system operator (ISO) or regional transmission operator (RTO) for cities located in regions with organized wholesale markets; '–' indicates absence of an organized wholesale market.
AR	Retail choice	In cities where the retail market is deregulated, this field indicates (where data is available) whether electricity choice is offered to consumers through the relevant utility suppliers for each city (Yes/No/Limited). Cities serviced by a municipal utility are assumed to have no retail choice. See the following column (AS) for details.
AS	Retail choice details	Offers additional qualitative information on the scope and location based eligibility for customers in areas with electricity choice offerings.
AT	Net metering	Indicates if a net-metering option is provided through the relevant utility supplier for each city (per DSIRE).
AU	3rd party PPAs	Categorization of state 3rd party power purchase agreement (PPA) legislation as authorized, disallowed or restricted, or permitted with system size limitations. Left blank if ambiguous.

Column	Field Title	Explanation
AV	State-level RPS	Identifies whether the state or region has implemented <i>mandatory</i> renewable portfolio standards (RPS) and if so, the relevant details on impacted cities and deadlines.
AW	Home rule state	Denotes whether the city is located in a Home Rule state (Yes/No) and thus the level of governing authority granted to municipalities. Cities in Home Rule states have more power over local decision-making (such as passing energy ordinances) than those in Dillon's rule states. Cases in which individual larger cities within Dillon's Rule states are granted Home Rule authority are also noted.
AX	CCA	Specifies if legislation in the state authorizing community choice aggregation (CCA) is under consideration (In progress) or has been passed (Yes/No) to enable local governments to aggregate the electricity loads of residents, businesses, and/or municipal facilities.
AY	Additional notes	Notes often pertaining to the electricity supplier or upcoming changes to relevant local regulatory structures or city agreements.
BA-BG	Primary Sources	Links to the raw documents and underlying sources for data collection.
	Contact information	Municipal sustainability or energy procurement representatives and contact information used by the author, Miela Mayer, in reaching out to conduct interviews where possible. **Note that this information cannot be published and thus will be removed from the final version, at the moment it is actually not in the excel file #1 but rather the targets tab of the excel file #2**

Note: (1) Specifically, California cities that are excluded, but that have formally adopted renewable energy targets, include, but may not be limited to: Chula Vista, Del Mar, Lancaster, Los Angeles, Menlo Park, Monterey, Nevada, Palo Alto, San Diego, San Francisco, San Jose, Santa Barbara, Solana Beach, South Lake Tahoe, and Truckee. (2) Denotes columns that are 'hidden' in the spreadsheet for readability. (3) Entities may have both achieved and in-progress targets, in these cases multiple statuses are given in consecutive order. (4) Entries are limited to the data reported by the EPA Green Power Purchasing program and NREL.

B.2. Excel file 2: Results, analysis, and data visualization tools

The following 'US Cities RE Targets – Results and underlying analysis' excel file #2 contains the final forecast upon which this analysis is based as well as the demand and supply fundamentals that feed into the forecast. In the results, this file also includes two useful data visualization tools: one for toggling by regulatory landscape and geographic factors, and another for toggling the demand forecast by city.

Table 9: Excel file #2 – Description of tabs

Tab (excel sheet)	Explanation
Cover	Cover page and table of contents
Results book – Net demand	The final, hardcoded, net-demand resulting from this analysis for each of the 61 'active' cities captured. This tab also contains hardcoded key information on deadlines and scope of commitments. All data presented here reflects the analysis conducted based on the 'Inputs' and 'Targets' tabs.
Tool 1 - Net demand by city	Data visualization tool #1: This tab contains the annual renewable demand forecast for all 61 active cities and a data visualization tool for interacting with the results. The graphs at the top depict the aggregate and net demand forecasts for a given entity. This tool can be toggled by selected city to generate custom net demand depictions for each of the 61 cities with active renewable energy targets captured in this analysis. To change the graph, choose an entity from the dropdown list in the orange user selection cell (E17). When the selected entity is on "All" the graph will depict the collective net demand forecast for all 61 cities (similar to the graph in Figure 4 of this paper). Users can also select from different REC purchasing assumptions in cell E19. The analysis conducted in this paper assumes cities will continue procuring renewable according to their historic purchasing trends 'historic trend'. This tool however gives the option to see how the net demand of these cities would change based on different percentages of renewable energy credit purchasing. The top half of this tab is frozen so that the graph and selection tool remain visible at all times. Rows 22-35 contain a dynamic data table upon which the graphs are based. Based on the entity and REC purchasing trend selected in the orange cells, this table pull the relevant numbers from the complete dataset below to alter the graph's depictions. The analysis portion of this tab is contained within the 'complete data set' (rows 37-814). This data is broken into different calculation 'Segments' or blocks: Each block contains 65 rows (one for each city, and four buffer rows for expanding the dataset) of a particular formula. The 'Segment' column of this data (column E) describes what calculation is occurring within each block. For some calculation blocks, the first row will contain a little note highlighted in light yellow to describe in more simple terms the purpose of the formula for that particular block. All data feeding into both tool 1 and tool 2 is pulled from the relevant supply, demand and inputs tabs.
Tool 2 - Net demand by criteria	Data visualization tool #2: Net renewable demand forecast for all active cities displayed in 5-year increments (2020, 2025, 2030, 2035, 2040). This data visualization tool can be toggled similar to tool #1 but in this tool, it is by selected criteria allowing users to filter by region, market, different regulatory structures, electricity supplier type, etc. to generate customized unmet demand graphs. To select desired criteria, choose from the dropdown lists in the orange cells (18O through 18W).
Demand	The Demand tab details the annual electricity and renewable demand forecast through 2040, including both active and looming cities captured in this analysis.
Supply	U.S. city and municipal Power Purchase Agreement and onsite generation inventory: This tab details the known renewable energy supply side of the story for different entities, capturing their contracted generation in each year between 2010 and 2030. See Table 10 below for a description of the fields within this tab.

Tab (excel sheet)	Explanation
	<i>Note: entries in this field are not all inclusive due to quality and availability of data</i>
Targets	Detailed underlying raw data with all 93 active, looming, or achieved renewable energy targets captured in this analysis. This list of targets additionally includes the California cities not covered by the descriptive dataset. While analysing the strategies for cities in California is outside the scope of this research due to the complexities surrounding new developments in CCAs, their demand presents equal opportunity for new development and thus they are included in this results analysis for the purposes of incorporating their demand profiles into the forecast. For a description of the fields within this target tab please refer to Table 7 in Appendix B.1 as this tab was made from the descriptive dataset and thus retains the same columns (as well as some additions for internal analysis). <i>Note that: the descriptions in the accompanying Dataset file are more detailed, refined, and updated than those within this targets tab as the targets tab is more for providing the internal raw data upon which the inputs tab is based than for the reader.</i>
Inputs	The inputs tab contains all the data that forms the basis of this analysis. The data in this tab was originally pulled from the raw data collected in the targets tab (essentially the first level of inputs). The inputs tab has been adjusted for this analysis and is now hardcoded so as to support building the other analysis tabs off of it.
List and state-level data	List of all captured cities and defined names for drop down menus, along with state RPS assumptions.
Maps	Maps of cities included in dataset, colored by status and sized by population, and the surrounding regulatory structures. These maps were generated using QGIS and are based on collected data. Location coordinates for each city are based off of Google Maps Data, regulatory information is consistent with the Database of State Incentives for Renewables & Efficiency (DSIRE), and population data for sizing the bubbles comes from the U.S. Census Bureau.
Figure 5	'Figure 5 in the accompanying research note: U.S. cities ranked by annual net renewable electricity demand, 2035

Table 10: Supply tab – Description of the fields

Column	Field title	Explanation
E	Type	PPA – offsite PPA Onsite – onsite generation (PPA or host-owned) Own – ownership (for example of a large biomass generator, etc.)
H	Tech	Identifies the renewable technology type of the resource facility: W – Wind PV – Solar O – Other, includes hydropower, landfill gas, biomass and geothermal.
I	Size	Volume of energy contracted or expected (MWh)
J	Sign date	Year in which the contract was first executed
K	Tenor	The lifetime of the renewable energy project beginning with the commercial operations date (start date). In the absence of data this length is assumed to be 20 years.
O	Capacity factor type	A – Actual capacity factor based on generation disclosed in official documentation E – Estimated capacity factor. All estimate values are based on the U.S. EIA 2016 annual averages: 35% Wind, 27% Solar PV, 38% Hydroelectric, 71% Landfill Gas and Municipal Solid Waste, 47% Biomass, and 75% Geothermal.
Q-AU	Annual generation	Estimated generation contracted or expected in each year between 2010 and 2040, in MWh

Column	Field title	Explanation
AW-AY	Links	Contains links to the original sources of each supply contract. <i>Note that these sources are often from bond documents in the Bloomberg terminal or Bloomberg New Energy Finance databases.</i>

Appendix C. City level data

Table 11 provides a simplified summary of each city's renewable target information. The following table includes full descriptions of each city's strategy and progress as of 2018. Please see the accompanying 'U.S. City renewable energy targets and strategies' Excel dataset—detailed in Appendix B—for a more granular look at all that's displayed below, as well as detailed regulatory and geographic information, and city specific sources.

C.1. Quantitative city data

Table 11: Renewable target and supply fundamentals of the 61 'active' cities

City	State	Final target (%)	Deadline ¹	Municipal operations only	Population	Targeted electricity demand (GWh, 2017)	Electricity demand CAGR (%)	Electricity supplier type	Retail electric choice ³
Phoenix	AZ	15%	2025	Yes	1,615,017	671	0.17%	IOU	No
Chula Vista	CA	100%	2035	No	267,172	1,778	0.44%	IOU	Yes
Goleta	CA	100%	2030	No	30,850	205	-0.95%	IOU	Yes
Lancaster	CA	100%	2020	No	160,106	1,065	0.34%	CCA	Yes
Menlo Park	CA	100%	2035, 2030	No	33,888	225	-0.95%	CCA	Yes
Monterey	CA	100%	2035*	No	28,454	189	-0.95%	CCA	Yes
San Diego	CA	100%	2035	No	1,406,630	9,360	0.44%	CCA	Yes
San Francisco	CA	100%	2030	No	870,887	1,187	0.66%	CCA	Yes
San Jose	CA	100%	2022	No	1,025,350	6,100	-0.95%	IOU	Yes
Santa Barbara	CA	100%	2030, 2020	No	91,930	612	0.78%	IOU	Yes
Solana Beach	CA	100%	2035	No	13,449	89	0.44%	IOU	Yes
South Lake Tahoe	CA	100%	2032	No	21,717	145	-0.59%	IOU	Yes
Truckee	CA	100%	2035, 2020	No	16,391	109	-0.94%	Political Subdivision	Yes
Boulder	CO	100%	2030	No	108,090	1,056	0.28%	IOU	No
Breckenridge	CO	100%	2035	No	4,896	48	0.28%	IOU	No
Colorado Springs	CO	20%	2020	No	465,101	4,516	0.03%	Muni	No
Longmont	CO	100%	2030	No	92,858	907	-0.47%	Muni	No
Pueblo	CO	100%	2035	No	110,291	1,077	1.10%	IOU	No

Orlando	FL	100%	2035*, 2030	No	277,173	3,168	1.26%	Muni	No
Sarasota	FL	100%	2035*, 2030	No	56,610	762	0.99%	IOU	No
St. Petersburg	FL	100%	2030	No	260,999	2,983	-0.19%	IOU	No
Winter Park	FL	60%	2030	No	30,208	345	1.61%	Muni	No
Atlanta	GA	100%	2035, 2025	No	472,522	6,227	-0.78%	IOU	No
Chicago	IL	100%	2025	Yes	2,704,958	1,800	-0.97%	IOU	Yes
Rockville	MD	100%	2022	Yes	66,940	10	-1.24%	IOU	Yes
Portland	ME	100%	2035	Yes	66,937	4	0.79%	IOU	Yes
Columbia	MO	30%	2029	No	120,612	1,213	-0.10%	Muni	No
St. Louis	MO	100%	2035	No	311,404	4,166	-1.36%	IOU	No
Grand Rapids	MI	100%	2025	Yes	196,445	168	0.12%	Muni	No
Traverse City	MI	100%	2020	Yes	15,287	12	-0.16%	Muni	No
Rochester	MN	100%	2031	No	114,011	1,225	-0.74%	Muni	No
St Cloud	MN	80%	2018	No	67,641	816	0.45%	Coop	No
Buncombe	NC	100%	2035*, 2030	No	256,088	3,378	-0.95%	IOU	No
Hillsborough	NC	100%	2035*	No	6,568	87	-1.03%	Coop	No
Hanover	NH	100%	2030	No	11,416	94	0.15%	IOU	Yes
Albuquerque	NM	25%	2025	Yes	559,277	20	-0.23%	IOU	No
County & Town of Taos	NM	100%	2030	No	33,065	367	-2.05%	Coop	No
Santa Fe	NM	100%	2025	Yes	83,875	931	-0.23%	IOU	No
East Hampton	NY	100%	2020	No	22,009	166	-0.45%	Muni	No
New York	NY	100%	2035	Yes	8,537,673	450	-0.36%	IOU	Yes
Ulster County	NY	100%	2035, 2017	Yes	179,225	1,352	-0.53%	IOU	Yes
Cincinnati	OH	100%	2035, 2018	No	298,800	3,839	-0.67%	IOU	No
Columbus	OH	31%	2030, 2019	No	860,090	783	-0.84%	Muni	No
Portland	OR	100%	2035*	No	639,863	1,875	0.68%	IOU	No
Multnomah	OR	100%	2035	No	159,903	361	0.68%	IOU	No
Downingtown	PA	100%	2035	No	7,905	90	-0.01%	IOU	Yes

Philadelphia	PA	100%	2035	No	1,567,872	14,650	-0.86%	IOU	Yes
Phoenixville	PA	100%	2035	No	16,885	193	-0.01%	IOU	Yes
Pittsburgh	PA	100%	2030	Yes	303,625	104	-0.88%	IOU	Yes
West Chester	PA	100%	2035	No	19,928	228	-0.01%	IOU	Yes
Columbia	SC	100%	2035	No	134,309	2,202	-0.66%	IOU	No
Austin	TX	65%	2027	No	947,890	12,825	1.34%	Muni	No
Denton	TX	70%	2019	No	133,808	1,458	1.63%	Muni	No
San Antonio	TX	40%	2035*	No	1,492,510	21,513	1.25%	Muni	No
Moab	UT	100%	2032, 2027	No	5,242	52	0.64%	IOU	No
Park City	UT	100%	2032, 2022	No	8,299	240	0.64%	IOU	No
Salt Lake City	UT	100%	2032, 2020	No	193,744	5,603	0.64%	IOU	No
Summit	UT	100%	2032	No	32,008	317	0.64%	IOU	No
Edmonds	WA	100%	2025	Yes	41,840	16	-0.81%	Political Subdivision	No
Mercer Island	WA	100%	2019	Yes	25,134	3	-0.30%	IOU	No
Madison	WI	100%	2035, 2030	No	252,551	3,002	-0.24%	IOU	No

Source: Local government websites, *EIA Form 861*, *DSIRE*, U.S. Census Bureau, *Bloomberg New Energy Finance*, and data collected from interviews. See Appendix A for details on methodology. 'Municipal operations only' refers to renewable targets that are set for just the municipal government's electricity demand (i.e., city owned buildings) rather than the entire city's electricity demand – see Appendix B.1 accompanying the *City Renewable Energy Targets and Strategies dataset* for further details on this distinction. More than one deadline year indicates an interim deadline either for municipal operations or for a smaller percentage of community-wide demand than that of the final deadline. (1) Deadlines marked with a star ("*") are assumed to be 2035 for the purposes of this analysis. (2) Indicates the target percentage was converted or adjusted from initially sourced figures reported in alternative units (i.e., in terms of GW capacity or GWh annual generation). (3) Field assumes that cities in municipal utility service territories do not have retail choice; see the field descriptions of the 'City Renewable Energy Targets and Strategies dataset' in Appendix B.1 for more details.

C.2. Descriptive city data

Tables 11 and 12 simply compile the qualitative columns (AF and AG) from the Active and Achieved tabs in the city targets dataset in order to provide a more concise overview of each city's strategy and documented actions to date.

Table 12: Summary of each cities' strategies and progress to date – includes 61 Active cities

Entity	General Strategy	Contracts signed or programs implemented to-date
1 Phoenix, AZ	No set strategy for achieving the community wide goal. Achieve 15% municipal electricity goal through renewable energy installations that are either city-owned or city-sponsored (primarily through public and private partnerships).	● RFI submitted seeking proposals for waste-to-energy information and technology to be utilized in the city. ● RFP submitted by APS for 10MW solar project to be built in partnership with the city (2015) ● 8.8MW offsite solar plant operational since 2015
<i>**12 CA cities excluded from descriptions due to complexities surrounding CCAs***</i>		
14 Boulder, CO	Actively pursuing municipalization (since 2010): filed applications to the Colorado Public Utilities Commission to establish a municipal electric utility and pass renewable energy offset requirements (2016). Plans to sign onsite PPAs to achieve half of its 100% target in the near-term.	● As of October 1, 2014, the city had 2,084kW of solar installed at city facilities with an additional 1MW under development. ● PPA for 1MW onsite solar commissioned 2010 ● City's targets are specified to account for increased demand: target of 100% community wide supply is set at 100MW by 2030 and 175MW 2050.
15 Breckenridge, CO	For the municipal-level commitment: work to sign long-term PPAs for renewables through Xcel Energy (local IOU) and pursue an additional onsite solar garden if it is cost effective. No set strategy for achieving the community wide goal; breckenridge is considering teaming up with other local counties to negotiate new options through Xcel Energy and pursuing community outreach programs.	● One on-site solar garden supplying municipal operations ● Xcel Energy is expected to release a SolarConnect program in the coming years allowing customers to pay extra for some renewable supply
16 Colorado Springs, CO	Exercise ability to work with municipal utility to influence community-wide renewable energy supply and set utility specific targets; sign long-term PPAs and pursue onsite generation.	● First utility scale solar plant became operational in 2016: CSU signed a 25 year PPA for the 10MW Clear Spring Ranch array ● City has 2MW of rooftop solar operational through the municipal utility's incentive program ● Colorado Springs Utilities is planning to add 50MW of wind energy
17 Longmont, CO	Work with local and wholesale municipal utilities (Longmont Power & Communications and Platte River Power Authority, respectively) to increase regional renewable electricity supply; count hydroelectric generation towards the renewable target.	● Longmont Power & Communication's energy mix is currently 19% hydroelectric power and 9.6% renewable ● Green Power Purchase Program through LPC: allows customers to increase the portion of renewable power in their electricity supply (up to 100%) by purchasing certified green power in set kWh blocks at an additional monthly subscription cost (\$3.12 per 100 kWh block for residents; \$15.5 per 500 kWh block for businesses).
18 Pueblo, CO	Considering exiting franchise agreement with Black Hills Energy in 2020 (requires purchase of natural gas power plant) and forming a municipal utility to purchase electricity directly on the spot market and/or pursue onsite generation. Target initiated by residential lobbying against high electric bills.	● Wind turbine located at the Vestas Pueblo facility produces electricity sold to Black Hills as RECs and feeds directly into the grid (operational since 2010). ● Vestas considering a 3MW wind project on site
19 Orlando, FL	Work closely with municipal utility (OUC) to achieve goal and maintain low electricity rates; collaborate with other communities in OUC service territory (includes Orange County) and seek partnerships on implementation pathways when most cost effective.	● Invested in solar on multiple public rooftops (includes 420KW array at Fleet & Facilities complex and 11KW array at Lake Eola) ● 5 MW onsite solar plant operational since 2016 ● "Solar Aggregation Program" newly launched by

		Develop detailed plan by 2019 (executed by Green Works Orlando team to include interim milestones, budget, financing mechanisms, and the percentage of renewable energy to be locally generated)	OUC to lower solar power costs through bulk purchasing (20-30% savings for residents and customers) ● City enabled Property Assessed Clean Energy (PACE) and Solar & Energy Loan Fund (SELF) financing programs (2016)
20	Sarasota, FL	No set strategy to date. Considering onsite generation to meet the municipal target; likely working with FPL on the community-wide target; looking into permitting and incentive options at the city level (leveraging home rule authority) and staying open to renegotiating/not renewing the 30-year franchise agreement with FPL if necessary.	● Two small-scale onsite solar projects exist at municipal sites ● FPL offers a Sunshine Energy program and a community-solar-like option to residents
21	St. Petersburg, FL	Allocate 250k of BP Oil Spill settlement funds to planning process; drafting an Integrated Sustainability Action Plan (ISAP) to roadmap meeting the renewable energy target (incorporates a CAP and strategies for the city's Net Zero target established in 2016).	● Recent Initiatives include: improve energy data collection and tracking; partner with the USF Clean Energy Research Center to conduct an initial analysis of immediate retrofits and projects; include solar PV designs in new construction of large capital projects; Biosolids Waste to Fuel Project (with grant from DOE). ● Solar projects installed at over 20 park sites and two high schools.
22	Winter Park, FL	No set deadline or strategy to date. Exercising ability to work with municipal utility to influence community-wide renewable energy supply.	● Sustainable energy plan adopted in 2015
23	Atlanta, GA	Publish plan for achieving target by Jan 2018 (to be written by the city's Office of Sustainability and include interim milestones, budget estimates, equity metrics, financing mechanisms, and the percent of renewable energy to be locally sourced or distributed generation).	● 1.5MW rooftop solar planned for construction on 24 city buildings (2017) ● Received SolSmart Gold designation from DOE for implementing programs and policy to facilitate solar growth (July 6th 2017) ● Launched a "Solar for All" initiative (2017) ● Developing a sustainability scorecard to track the progress of all projects across the various government departments.
24	Chicago, IL	Sign long-term PPAs with utilities, increase onsite generation, and purchase short term RECs generated under the Illinois RPS to meet the demand of ~900 municipally owned buildings.	● Installed 900+ solar panels on rooftop of Shedd Aquarium ● City eliminated coal from its annual energy purchases in 2013 (over 1 billion kWh) ● Energy benchmarking ordinance requires buildings >50,000 square feet to report energy use annually (and verify data every 3 years) ● Smart Lighting Project and multiple other efficiency measures implemented community-wide
25	Rockville, MD	Purchase short-term wind RECs covering approximately 100% of Rockville Municipal Operation (achieved). Meet 20% of municipal operations electricity demand (2,030MWh) with on-site small-scale solar and geothermal generation.	● Intent announced in 2013 to Partner with the Maryland Energy Administration, and to Enroll as a Maryland Smart Energy Community
26	Portland, ME	Amend and revamp the 2008 Municipal Climate Action Plan to include a strategy to achieve the 2040 goal (action planned to be completed by the Portland sustainability committee).	● PPA signed jointly with South Portland, ME for each city to install 660kW solar on their respective closed landfills
27	Columbia, MO	Exercise ability to work with municipal utility to influence community-wide renewable energy supply: Implementation plan is to generate or purchase electricity generated from eligible renewable energy sources at the following levels (% of electric retail): 2%	● PPA for city's cheapest wind so far (\$21/MWh) approved 06/07 - 35MW potentially operational in 2021 (*pending Missouri PSC's approval for construction of Grain Belt Express Clean Line, as it is the main means of transporting this energy).

		by 2008, 5% by 2013, 15% by 2018, 25% by 2023 and 30% by 2029.	• On track to meet the 15% RPS by 2018 according to Water and Light Director • City adopted 2004 ordinance that mandates Columbia Water & Light to purchase increasing amounts of renewable energy - utility is required to submit a plan annually outlining compliance (true as of 2017)
28	St. Louis, MO	No set strategy to date. City is in the process of developing a plan to wean off fossil fuels by December 2018.	• Just before St. Louis approved their renewable energy target, the city's primary utility, Union Electric Co (subsidiary to Ameren Corp), announced intentions to invest \$1 billion in bringing at least 700MW of wind capacity online by 2020. This could lead renewables to account for about 10% of the generation mix supplied to St. Louis in coming years.
29	Grand Rapids, MI	Exercise ability to work with municipal utility to influence renewable energy supply; investigate and pursue onsite generation projects (considering development of biosolids biodigesters, additional solar panel projects, and geothermal projects).	• RFP submitted Jan 2015 for 15-20 year contract to develop 4MW solar project on the Butterworth Landfill site (could cover 40% municipal demand) • Developed special tariff (approved by Michigan PSC) allowing city to enrol in the Green Generation Program • City owns a 135kW onsite solar project (2012)
30	Traverse City, MI	Sign long-term PPAs for both small and utility-scale renewable energy projects. Exercise ability to work with municipal utility to influence renewable energy supply: TCL&P works with the Michigan Public Power Agency to pursue larger scale projects.	• PPA signed for 3.6MW from offsite utility scale wind project (in Michigan's Thumb region)
31	Rochester, MN	Exercise ability to work with municipal utility to shift RPU electric generation from coal and fuel oil to a mix of natural gas, fuel oil, hydroelectric generation, and solar PV.	• RPU utility not renewing contract with Southern Minnesota Municipal Power Authority in 2030 in order to be coal free and allow for more renewable energy sources • Report in 2016 identified need for full time sustainability coordinator to act as a liaison for Rochester Energy Commission and ensure the action plan is executed (status TBD) • City Council created an Energy Commission in 2009 to develop, implement and track a Local Energy Action Plan • Rochester enrolled in the Minnesota Green Step Cities program in 2013
32	St Cloud, MN	Focus on onsite generation, work with Xcel to enable community solar, and sign long-term fixed price PPAs. Existing hydroelectric generation is excluded for purposes of additionality. City set 80% as a general target for reducing reliance on fossil fired sources (so that strategies may include various measures such as energy efficiency or REC offsets in addition to transitioning to renewable energy). However, city is prioritizing new renewable energy and as of 2017 is on track to exceed the (ambiguous) 80% target by about 10%.	• RFP submitted for the development, design, financing and construction of PV facilities at multiple city locations (May 2015). Proposals may be for both roof and ground-mount locations, multiple PPAs may be awarded under the RFP. • RFP resulted in the development and ongoing implementation of 27 community solar gardens (limited by the Minnesota PUC to 5MW of capacity each). As of 2017, the city has subscribed to about 54MW with 25-year fixed rate PPAs (agreement between Xcel and the state PUC limits subscriptions to 40% of each garden). • 7 behind-the-meter rooftop solar arrays planned for installation on city facilities (totalling 754kW capacity, net-metered, procured through PPAs). • The largest 3 of the 7 (134kW, 200kW and 220kW) will be online by the end of 2017.
33	Buncombe, NC	Pursue onsite generation and purchase renewable electricity through Duke Energy's Green Source Rider program to offset municipal operations. No set	• City installed onsite solar on the rooftop of their municipally owned Human Services building and has plans to lease land to Duke Energy for an additional

		strategy for meeting the community wide goal; Buncombe increased their original 80% target to 100% after Duke Energy announced its recent 17% increase in electricity rates - the county hopes to save on electricity costs community-wide by switching to renewable energy.	5MW onsite solar project. ● Green Source Rider Program allows large customers (i.e. municipalities, universities, and the local military base) in North Carolina state to sign power purchase agreements for renewable electricity through Duke Energy.
34	Hillsborough, NC	No set strategy to date. Hillsborough has intentions to combine efforts with its surrounding county, Orange NC, and neighbouring communities to source renewable generation and increase the community-wide options available.	● Green Source Rider Program allows large customers (i.e. municipalities, universities, and the local military base) in North Carolina state to sign power purchase agreements for renewable electricity through Duke Energy.
35	Hanover, NH	Purchase short-term RECs to meet the majority of the community-wide demand and pursue large scale community solar arrays to supply municipal operations; work with Dartmouth College on planning and implementation.	● Solarize Hanover Program - increases community's renewable power generation (62 additional residential solar arrays) ● City negotiated with utility to offer a 100% green-e certified residential power rate to interested residents and employees ● Capture and use of methane implemented at the Hanover Water Reclamation Facility
36	Albuquerque, NM	Pursue onsite solar generation at city facility locations; Department of Municipal Development heads all progress and is encouraged to consider cost effective clean energy options on an ongoing basis.	● Taking the first steps to install over \$25 million worth of solar projects on municipal buildings (Currently municipality generates 3% of energy from solar) ● 1MW onsite solar plant (located at Albuquerque International Sunport) signed 2010 and operational since 2012. ● City ranked 9th for amount of solar installed community-wide in 2016 (82MW)
37	County & Town of Taos, NM	Joint effort between cities and towns in the Kit Carson Electric Coop (KCEC) service area, led by Renewable Taos, inc. Strategy: leverage the June 18, 2015 FERC order to terminate contract requiring KCEC buy 95% of its energy through 2040 from Tri-State Generation and Transmission Association. Post contract exit: develop local PV generation sold directly to KCEC and sign long-term PPAs for wind to cover 90% of peak demand by 2020.	—
38	Santa Fe, NM	No set strategy to date; Santa Fe is in the planning stages and has approved a resolution directing the City Manager to develop a Feasibility Study for working with the Public Service Co of New Mexico to meet their 100% goal over the next 90 days. The renewable target is part of a larger carbon neutrality goal for which the city initiated a 25-year sustainability plan in 2015.	● About 25% of the city's energy use comes from renewable resources to date, primarily onsite solar installations on municipal structures — including the Genoveva Chavez Community Center, the Buckman Direct Diversion Project, the Community Convention Center and wastewater treatment plant — collectively 4.8 MW of renewable capacity.
39	East Hampton, NY	Work with consultant on renewable energy policy options; consider rebate and tax incentives, expedite solar and wind permitting for commercial properties; pursue onsite generation.	● Both local utilities have strong solar Feed-In-Tariffs ● Several utility-scale solar farms planned for town-owned land: 10 potential sites identified, 3 of which are planned to produce 4.8MW of power. ● 90MW offshore wind farm announced 2017 will deliver a portion of its generation to East Hampton ● NY state committed to developing 2.4GW of offshore wind in the same area by 2030 ● 2.3MW onsite solar project operational since 2012 (developed and operated by Borrego Solar Systems)
40	New York, NY	Leverage purchasing power (as one of the largest energy purchasers in the country) to work with utility to drive development of new renewable energy sources	● RFI submitted July 2015 to identify new sources of renewable energy and potential projects to supply government operations with 100% clean electricity.

		through independent developers and to enact policies that influence the energy supply portfolio; submit RFI on ideas for cost-effective options. governor Cuomo Clean Energy Fund. Reforming the Energy Vision. NY state committing money from REGGI to NYGB to accelerate the deployment of RE with financial support. 100 percent of City government operations from renewable sources of energy.	● Also seeking information on financial mechanisms to support renewables (like green bonds).
41	Ulster County, NY	Achieved coverage of 150% municipal demand in 2016 with RECs, now pursuing increase in offsite generation through long-term PPAs to cover upwards of 25% of community wide demand.	● PPA for 2.53MW offsite solar project (commenced construction fall, 2016) ● Onsite 3.41MWh solar project (ground mounted at former landfill in the Town of Ulster)
42	Cincinnati, OH	Contract cost effective (i.e. cheaper than current utility rates) green energy from Dynergy Inc., for municipal facilities (police and fire stations, health clinics, recreation centers, and all city administration buildings) starting January 2018 through at least 2021; purchase short-term voluntary RECs to meet the rest of their municipal electric load.	● City signed an electricity contract with Dynergy Inc., to purchase 100% green energy for almost all city facilities beginning January 2018. Intentions to power buildings with renewable energy through at least 2021 ● Cincinnati has been aggressive in its push for renewable energy; since 2012, the Cincinnati Electric Aggregation Program has provided 100% green electricity for 80,000 households and is currently the fourth-largest buyer of residential green energy in the county ● Cincinnati's Mayor committed to install 25 MW of onsite solar across the city ● Solarize Cincinnati initiative, a new program sponsored by the City, makes purchasing solar panels for private homes simpler and more affordable
43	Columbus, OH	Exercise ability to work with municipal utility to influence renewable energy supply; pursue onsite generation and support residential renewable energy programs; heavy reliance on RECs.	● Department of Public Utilities involved in school and restaurant rooftop solar instalments (2011), parking meters have been replaced with solar operated meters. ● 5MW O'shannessy Dam (operational since 1987) ● Support and permitting information provided to residents interested in installing solar or micro-wind
44	Portland, ME	Amend and revamp the 2008 Municipal Climate Action Plan to include a strategy to achieve the 2040 goal (action planned to be completed by the Portland sustainability committee).	● PPA signed jointly with South Portland, ME for each city to install 660kW solar on their respective closed landfills
45	Multnomah, OR	Work closely with Portland, OR to employ similar tactics. *In order to avoid double counting, the figures reported here for Multnomah County exclude Portland City.	● Local utility (PGE) has an RPS of 15% and their coal-fired plant is expected to close in 2020
46	Downingtown, PA	No set strategy to date; Downingtown Borough just announced on the 5th of December, 2017 their goal of powering all electricity across the community with 100 – percent clean energy by 2035 and all heat and transportation by 2050.	
47	Philadelphia, PA	Pursue on and offsite renewable energy (particularly solar); investigate feasibility of increasing onsite cost effective generation with RFI; utilize RECs if necessary.	● 2050kW city-owned solar array at the Southeast Water Pollution Control Plant (2011, about 340MWh /year produced) ● 5.6MW Biogas Co-generation facility operational as of December 2013. (28,000MWh/year) ● 3.8MW of an expected 57.7MW of solar by 2021 is currently online (2016) ● Request for information released regarding project structures that make renewable energy development projects feasible for city

48	Phoenixville, PA	No set strategy to date though the borough's resolution calls for prioritizing onsite generation (which may include small-scale hydropower); Phoenixville's resolutions followed day's after Downingtown PA's December, 2017 announcement of a similar goal.	–
49	Pittsburgh, PA	No set strategy to date though some onsite generation is under consideration and heavy reliance on REC purchases is likely.	• 10% of the Pittsburgh's municipal electricity comes from a wind PPA and 30% is offset by REC purchases • Developing city's third climate action plan (PCAP 3.0) • Wind study is being completed along the Allegheny River to determine feasibility of installations on the North Shore
50	West Chester, PA	West Chester Borough currently purchases green power to cover its entire municipal electric load; no set strategy to date for meeting the community-wide target.	–
51	Columbia, MO	Exercise ability to work with municipal utility to influence community-wide renewable energy supply: Implementation plan is to generate or purchase electricity generated from eligible renewable energy sources at the following levels (% of electric retail): 2% by 2008, 5% by 2013, 15% by 2018, 25% by 2023 and 30% by 2029.	• PPA for city's cheapest wind so far (\$21/MWh) approved 06/07 - 35MW potentially operational in 2021 (*pending Missouri PSC's approval for construction of Grain Belt Express Clean Line, as it is the main means of transporting this energy). • On track to meet the 15% RPS by 2018 according to Water and Light Director • City adopted 2004 ordinance that mandates Columbia Water & Light to purchase increasing amounts of renewable energy - utility is required to submit a plan annually outlining compliance (true as of 2017)
52	Austin, TX	Sign long-term PPAs with offsite utility-scale (wind, solar) projects supplemented by municipally owned generation (landfill gas, biomass) and smaller-scale community solar programs.	• City met 2007 goal of 100% renewable energy for municipal operations in 2011 • PPAs for 595.5MW of utility-scale solar of which 275.5MW is operational • Issued an RFP for an additional 150MW of utility-scale solar in May 2017 • PPAs with >1 GW of wind, 100MW of biomass and 8.2MW of landfill methane
53	Denton, TX	Sign long-term PPAs for offsite wind (~52%) and solar (~17%) projects; retire coal and develop quick-start natural gas plants to hedge against the variability of renewables; sell excess energy into the ERCOT wholesale market. Purchase short-term RECs to fulfill remaining renewable demand.	• Signing long-term PPAs for about 500MW of renewables (expected peak load by 2035) • PPA for 30MW solar scheduled to start in 2019 • RFP submitted seeking PPA for 200MW wind project (July, 2016)
54	San Antonio, TX	Sign long-term PPAs with offsite utility-scale (wind, solar) projects; supplement with smaller-scale community distributed generation programs (solar, landfill gas, micro-wind).	• Currently 1059.1MW of operational wind through PPAs (2016) • Currently 446MW of operational solar through PPAs (2016) • "agreement in place for additional solar power" (2017) • CPS Energy submitted RFP to develop a PV system for a microgrid installed at Fort Sam Houston in San Antonio, Texas. (2015) RFP also submitted for rooftop solar development in San Antonio.
55	Moab, UT	Sign long-term PPA's with RMP Blue Sky; pursue onsite solar at municipal facilities; promote residential renewable energy through programs with IOU.	• 63kW onsite solar array (city hall building) • Moab residents can participate in RMP Solar Subscriber program (*90% full) - PPA signed for 20MW

			solar in Millard County, UT operational as of 2017, customers subscribe in blocks of 200kWh
56	Park City, UT	Work collaboratively with local utility (leveraged ability to pursue paths outside of RMP (municipalization, CCA) to get IOU to sign Joint Clean Energy Cooperative agreement); set targets not just for renewable energy but also to be net-zero carbon by deadline; define 100% renewable on annualized basis as entirely new build – excludes any existing or planned renewable energy through RMP to date as well as existing unbundled RECs/grid renewables; sign long term PPAs for utility-scale offsite wind and solar; pursue onsite generation for municipal operations when cost effective; release public documentation of agreements and updates on an annual basis.	• Approved net zero building standard for municipal facilities (first ever for any) • Currently no paths exist for CCA in Utah but joint agreement commits RMP to providing similar options under the premise that the city does not seek programs elsewhere • Actively conducting energy procurement study analysis: considering RFP submission and potential to sign joint PPAs with SLC through RMP. • Residential renewable energy systems eligible for building fee waiver (ABCs Permitting program). Net metering programs for rooftop solar are under consideration.
57	Salt Lake City, UT	To meet municipal operations goal: Sign long-term PPAs, install NEM projects, Subscribe to RMP utilities Solar program, submit RFP for large scale renewable energy projects, provide Climate Positive SLC website to track progress. To meet community-wide target: Work collaboratively with local utility (leveraged ability to pursue paths outside of RMP (municipalization, CCA) and pre-set renewable targets to get IOU on board); expedite the solar permitting process, optimize interconnection for distributed generation, promote large-scale clean energy choices for residents through RMP.	• City signed Joint Clean Energy Cooperative agreement with RMP • SLC has 3MW onsite generation (distributed among 12 municipal installations) • RMP offers a Solar Subscriber program to SLC residents - PPA signed for 20MW solar in Millard County, UT operational as of 2017, customers subscribe in blocks of 200kWh • Collaborated with RMP to start Blue Sky renewable energy program - enrolment leads to REC purchasing from newly developed facilities. • PPA for 1.65MW onsite solar commissioned 2012 • SLC released "Climate Positive 2040" plan specifying additional steps and policies the city will pursue with RMP to achieve the 2032 city wide goal
58	Summit, UT	-	-
59	Edmonds, WA	No set strategy to date	• Expansive climate and sustainability information and resources available on city's website (documents a range of energy related initiatives since 2006) • Published Energy Action Plan in 2011 focusing on efficiency and distributed generation - plan includes developing multiple incentives and requirements to promote participation in renewable energy programs • City joined the New Energy Cities program in 2010 and has been working on a 20 year clean energy strategy
60	Mercer Island, WA	Sign long-term PPAs for utility scale wind projects through PSE to cover entire demand.	• City participates in PSE Green Direct program allowing Washington cities to elect for direct wind energy purchasing • 20-year PPA for wind power (Operations intended to begin in 2019 and cover 100% municipal demand) • Ran successful residential Green Power sign-up program in 2012 • Community-wide solar generation is approaching 1,000kW of capacity (103 locations)
61	Madison, WI	No specific dates set but allocated \$250,000 to 2017 planning process and working with a consultant to develop strategies for meeting targets on an aggressive and medium timeline (chosen between by city council).	• As of 2017, MG&E and city officials are negotiating an updated Memorandum of Understanding (MOU) to enable joint clean energy projects (plans for expanding solar, electric vehicles, and grid modernization for meeting the community wide goal). MOU is result of Energy Work Plan adopted in 2016

	Publishing plan with benchmarks and strategy for municipal operations by Jan 2018; subsequently address community-wide transition. City set net-zero carbon target which gives more flexibility in how they go about meeting 100%. City operations are supplied entirely by MG&E and will thus be 30% renewable by 2030, for the other 60% considering: onsite generation (both through 3rd party PPAs with NEM, and entirely behind the meter); procuring new generation through MG&E Energy Rider program; converting to an electric fleet and building HVAC system to lower reliance on natural gas; and purchasing RECs (prioritizing additionality) in the short term.	• New Energy Rider program approved (2017) - allowing commercial and government customers to meter energy use and apply to work with MG&E to develop new renewable energy projects that can supply the metered demand - may be onsite or offsite, ensures additionality • Green power tomorrow program through MG&E: lets customers purchase power from wind and solar (solar program sold out) by agreeing to pay more per kWh. All generation in this plan however is existing renewable energy that MG&E needed in order to meet standards. The municipality of Madison thus does not participate in program as it ensures no additionality • Energy Cap program: web based software that tracks all electric meters for the city operations
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Table 13: Summary of each cities' strategies and progress to date – 17 Achieved cities

	Entity	General Strategy	Contracts signed or programs implemented to-date
1	Kodiak Island, AK	Meet demand with local hydro and wind, backed by a battery storage system. Utilize 3-5% diesel as needed but replace most with renewable generation. Progress facilitated by state grants and CREB bond issuance.	• 4.5MW wind onsite (2009) • Additional 4.5MW wind with 3MW battery storage added 2012
		1 CA city excluded from descriptions due to complexities surrounding CCAs	
3	Aspen, CO	Utilize onsite hydro capacity; sign long-term PPA's for utility-scale wind and solar; negotiate with Municipal Energy Agency of Nebraska; encourage HCE cooperative utility to increase renewable energy portfolio from 7% to 25%; include renewable energy resource requirements in utility franchise agreements. Aspen's current energy portfolio is ~50% wind, 45% hydropower, and 5% solar and landfill gas.	• Renewable incentive program for residential installations • Owns 2 hydroelectric plants (5MW and 450KW) controlled by Bureau of Reclamation. Signed PPA to meet 14% of community demand with hydropower (operational 2015); several micro-hydro projects.
4	Greensburg, KS	Pursue onsite generation, promote net metering and some small scale distributed renewable energy in residential locations (makes up about 8% of supply). Work with John Deere renewable energy and the Kansas Power Pool to build utility-scale wind farm onsite.	• 12.5MW Greensburg Wind Farm operational (enabled target achievement) • City replaced all distribution lines post 2007 tornado • Net Metering policy promoted residential RE
5	Boston, MA	Primarily focused on energy efficiency but purchases short term voluntary RECs and has a history of promoting onsite solar installations throughout the city.	• 2007 green power purchasing goal of 11% and 15% municipal operations by 2012 (met) • Boston Green Ribbon Commission (GRC) started renewable energy leadership prize and incentivises private sector investment in new projects • Solar Boston Program: goal to increase installed solar from 500KW (2008) to 25MW (2015): committed in 2008 to install \$1 million+ of solar thermal and PV on four municipal buildings by 2009.
6	Worcester, MA	First purchased short term RECs (achieved 100% of municipal operations through purchases), now actively working to increase onsite renewable generation and play a roll in helping MA meet its RPS for local RE.	• City has an NEM agreement with local solar farm to encourage residents and businesses to choose the clean energy option on their municipal utility bills. • MA State has a deregulated utility system

7	Brookeville, MD	Unspecified deadlines, but committed to offsetting energy use with renewable power. 100% municipal target achieved, and now working towards the community wide target.	• Town currently meets 50% of community-wide energy demand through REC purchases from Good Energy and WGL Energy
8	Columbia, MD	Exercise ability to work with municipal utility to influence community-wide renewable energy supply: Implementation plan is to generate or purchase electricity generated from eligible renewable energy sources at the following levels (% of electric retail): 2% by 2008, 5% by 2013, 15% by 2018, 25% by 2023 and 30% by 2029.	• PPA for city's cheapest wind so far (\$21/MWh) approved 06/07 - 35MW potentially operational in 2021 (*pending Missouri PSC's approval for construction of Grain Belt Express Clean Line, as it is the main means of transporting this energy). • On track to meet the 15% RPS by 2018 according to Water and Light Director • City adopted 2004 ordinance that mandates Columbia Water & Light to purchase increasing amounts of renewable energy - utility is required to submit a plan annually outlining compliance (true as of 2017)
9	Gaithersburg, MD	Purchase short term RECs from wind energy to offset 100% of electricity use from municipal operations	—
10	Ann Arbor, MI	City's resolution encourages meeting the goal with long-term, fixed-rate contracts with in-state (offsite) wind generation.	• City generates and uses about 8,900MWh annually from renewable energy sources, (primarily biogas, solar, and small hydroelectric facilities). • 20% municipal interim target achieved, city is now working to expand to community-wide operations but stated that "progress at the state is necessary to allow for a larger share of renewable energy to power homes, business, and industry."
11	Cologne, MN	Sign long-term PPAs with utility community solar program (SunShare)	• Developed SunSheare solar gardens with Xcel Energy's Solar Rewards Community Program • The state of MN passed Solar Energy Jobs Act making community solar possible (2013)
12	Rock Port, MO	Generate energy in excess of peak load from 100% onsite wind power; sell excess electricity to Missouri Joint Municipal Utilities.	• Four 1.2MW wind turbines within the city limits connect directly to the city's distribution lines.
13	Las Vegas, NV	Partner with NV Energy (IOU) to supply 100% municipal demand (~140 buildings) with renewable energy from a local solar plant: included signing long-term PPAs with utility, increasing onsite solar generation, and purchasing short term RECs.	• 6.5MW rooftop solar installation at Mandalay Bay Convention Center (NRG Energy) • 100MW Nevada Solar One facility (offsite, near Boulder) operational since December 2016 • Vegas contracted with NV energy to provide GreenEnergy Rider program in support of renewable initiatives
14	Ithaca, NY	Purchase RECs on a short-term basis to meet entire target.	• Two-year PPA with Integrys Energy Services of New York Inc.
15	Watervliet, NY	Sign long term PPAs for onsite solar; pursue onsite hydroelectric generation; purchase short-term RECs.	• Installed over 100KW of solar panels, including a 17KW solar system on the Hudson Shores Park boathouse and two larger 37KW PV systems on the Firehouse and the Senior Center • 1.25MW hydro power generation unit at the Watervliet Reservoir operational since 2008 and 5MW hydro plant operational since 2012
16	Dallas, TX	Purchase RECs from retail electric provider on a short-term basis.	—

17	Georgetown, TX	Sign long-term PPAs with offsite utility-scale (wind, solar) projects in excess of load requirements; sell excess energy into the ERCOT wholesale market.	● PPA for 154MW of utility-scale solar (2018) ● PPA for 144MW of wind (2015)
18	Burlington, VT	Procure onsite generation, sign PPAs for utility-scale offsite wind, purchase short-term RECs. Current renewable portfolio is 60% onsite biomass, 17% offsite wind, along with other small scale hydro and solar sources.	● Owns 7.4MW Winooski One Hydroelectric Facility onsite (located at edge of the city, operational 2014) ● RFP submitted jointly between UVM and Burlington City for 1MW of solar projects on UVM campus (2015)

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City-specific sources:

The following list includes an incomplete portion of the many city-specific sources used in compiling this analysis. Please see the compilation of all city-specific links to primary sources columns 'BG through BM within the 'Targets' tab of the second excel file: 'US Cities RE Targets – Results and Underlying Analysis'. (Source links are also available in excel file #1: 'US City Renewable Energy Targets and Strategies Dataset').

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